

Bengali Hate Speech Detection: A Multi-Track Benchmark with Fusion, Explainability and Uncertainty Quantification

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A thesis submitted in group fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering



Department of Computer Science and Engineering

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Declaration

This is to certify that the thesis work entitled **Bengali Hate Speech Detection: A Multi-Track Benchmark with Fusion, Explainability and Uncertainty Quantification** has been carried out by **Dipta Sarker**, Student ID: 12221125, **Md. Rahmatullah Mia**, Student ID: 12221100, **Mahdi Hasan**, Student ID: 12221067, and **M. Shawan Biswas**, Student ID: 12221096 in the Department of Computer Science and Engineering, University of Global Village (UGV), Barishal, Bangladesh. The thesis work, or any portion of it, has never been submitted to a university or other institution for the granting of a degree or diploma.

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Abstract

Bengali is spoken by over 230 million people, yet automated moderation tools for abusive and bullying content remain comparatively underdeveloped relative to English. This thesis presents a large-scale study of cyberbullying and hate-speech detection using 29,829 cleaned Bengali social-media comments across seven categories. Seventeen models spanning five tracks are benchmarked: classical TF-IDF models, static embeddings (Word2Vec, FastText, GloVe), deep sequence models (TextCNN, BiLSTM with attention), fine-tuned transformers (BanglaBERT, BanglaELECTRA), and fusion/stacked ensembles. All models follow an identical stratified 70/15/15 protocol and are evaluated using eleven metrics, bootstrap 95% confidence intervals, and pairwise McNemar tests. A stacked ensemble achieves the best performance (F1-macro = 0.8951, accuracy = 0.9068, MCC = 0.790), ahead of BanglaELECTRA (F1-macro = 0.8927) and TF-IDF + XGBoost (0.8828) by a small margin. The thesis also reports SHAP/LIME explainability, attention patterns, MC-Dropout uncertainty, deep-ensemble disagreement, and temperature scaling, which reduces Expected Calibration Error from 0.0324 to 0.0248. Data audits and ablations find no evidence of leakage or label-parsing artefacts and confirm benefits from character n-grams and tuned thresholds. To assess generalisability, the protocol is replicated on an independent BD-SHS corpus of 50,264 cleaned comments; the stacked ensemble again performs best (F1-macro = 0.9197), while rankings are nearly identical across corpora (Spearman $\rho = 0.988$, $p < 0.001$), indicating the findings generalise beyond the primary corpus. The work is motivated by the need for reliable moderation tools for Bengali online communities.

Keywords: Bangla hate speech, Bengali NLP, cyberbullying detection, low-resource language, transformers, BanglaBERT, BanglaELECTRA, TF-IDF, BiLSTM attention, model fusion, ensemble learning, explainable AI, SHAP, LIME, uncertainty quantification, calibration.

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List of Abbreviations

BERT	Bidirectional Encoder Representations from Transformers
BoW	Bag of Words
CI	Confidence Interval
CLS	Classification token (special transformer input token)
CNN	Convolutional Neural Network
ECE	Expected Calibration Error
ELECTRA	Efficiently Learning an Encoder that Classifies Token Replacements Accurately
F1-macro	Unweighted mean of per-class F1 scores
FN / FP / TN / TP	False Negative / False Positive / True Negative / True Positive
GRU	Gated Recurrent Unit
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory network
MC-Dropout	Monte-Carlo Dropout
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
PR-AUC	Area Under the Precision-Recall Curve
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations
SVD	Singular Value Decomposition
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
MCC	Matthews Correlation Coefficient

Chapter I

Introduction

1.1 Background and Motivation

The growth of Bengali language social media use across Bangladesh and West Bengal has been accompanied by a rise in cyberbullying, harassment, and hate speech directed at public figures, celebrities, and ordinary users, as reflected in the large volume of YouTube and Facebook comments collected and annotated for this purpose [1]. Because Bengali is a morphologically rich, low-resource language for NLP tooling, moderation systems trained primarily on English transfer poorly to it. Early studies on Bengali text classification have explored multichannel convolutional LSTM architectures to address the challenges of under-resourced Bengali language processing [2]. Efforts to make such systems more transparent have incorporated explainability techniques such as sensitivity analysis and layer-wise relevance propagation to interpret individual predictions [3], while larger and more diverse benchmark corpora have subsequently been released to broaden coverage beyond a single dataset [4], and hybrid transformer-based models have been proposed to handle Bangla hate speech across a range of social contexts [5].

Left unmoderated, abusive online content is not merely a nuisance to individual users; it can carry real psychological and social costs for victims, ranging from anxiety and withdrawal to, in severe cases, a link between sustained online harassment and offline harm. Automatically identifying such content early is of direct value to platform moderators and, in the most serious cases, agencies seeking to prevent escalating aggressive behaviour from resulting in real-world harm. This makes automated, reliable, and explainable Bengali hate-speech detection a problem with both a clear technical challenge and a clear social motivation.

We chose this research problem because cyberbullying and hate speech are increasingly visible in Bengali online communities, while reliable automated detection systems for Bengali remain comparatively limited. Through this work, we aim to contribute to the development of more accurate, trustworthy, and explainable approaches for identifying harmful online content. Prior work has typically evaluated a single model family, classical machine learning, a

recurrent network, or one fine-tuned transformer on one dataset, which makes it difficult to judge whether reported gains come from the representation, the classifier, or incidental hyperparameter choices. A second, related gap is that most reported Bengali hate-speech systems present a single accuracy number without accompanying uncertainty estimates, statistical significance testing, or an explanation of what the model actually attends to, which limits their trustworthiness for real deployment. This thesis addresses both gaps with a unified, multi-track comparative study conducted under one evaluation harness, one train/validation/test split, and one metric registry, so that classical, static embedding, deep learning, transformer, and fusion/ensemble approaches can be compared on genuinely equal footing, and so that the strongest model’s behaviour can be inspected rather than taken on faith.

1.2 Problem Statement

Given a Bengali social-media comment drawn from one of several topical categories (sports, entertainment, religion, politics, crime, celebrity, and memes/general content), the task addressed in this thesis is binary classification: determine whether the comment constitutes bullying/hate speech (label = 1) or non-bullying content (label = 0). Formally, given a comment x and a trained classifier f , the goal is to learn f such that $f(x)$ approximates the true label y in a way that generalises to unseen posts, while also

- Remaining computationally practical for the different representation families considered.
- Exposing a usable confidence signal for its own predictions.

The problem is complicated by:

- Morphological richness and comparative scarcity of pretrained Bengali NLP resources.
- An imbalanced label distribution (approximately 33% positive).
- Short, noisy, code-mixed (Bangla–English ”Banglish”) text typical of social media.
- Deliberately obfuscated profanity that evades naive word-level matching.

1.3 Objectives of the Study

- To build a controlled, multi-track benchmark comparing classical lexical models, static word embeddings, deep neural sequence models, and fine-tuned Bengali transformers for cyberbullying/hate-speech detection on a common dataset and evaluation protocol.
- To investigate whether combining representations across tracks via early (feature-level) fusion, late (decision-level) fusion, or stacked/soft-voting ensembles yields a statistically meaningful improvement over the best individually trained model.
- To evaluate the explainability of the trained models using global (SHAP, linear coefficients) and local (LIME) attribution methods, and to compare the attention behaviour of a transformer against a recurrent (BiLSTM) architecture.
- To quantify predictive uncertainty and calibration (MC-Dropout, deep-ensemble disagreement, temperature scaling) so that model confidence can support a human-in-the-loop moderation workflow.
- To audit the dataset and ablate key modelling choices (TF-IDF vs. bag-of-words weighting, decision-threshold tuning, auxiliary category features) to confirm that the reported performance reflects genuine task difficulty rather than a pipeline artefact.

1.4 Contributions

- A controlled, seventeen-model comparative benchmark spanning classical, static-embedding, deep-learning, transformer, fusion, and ensemble tracks, evaluated with eleven metrics, bootstrap confidence intervals, and pairwise McNemar significance tests.
- A stacked-generalisation ensemble and a validation-weighted late-fusion scheme that both outperform every individual model and every early-fusion variant on held-out F1-macro.
- A four-view explainability analysis (global linear/SHAP importance, local LIME case studies across the confusion-matrix quadrants, and transformer-vs-BiLSTM attention comparison) grounded in the same trained models used for the accuracy benchmark.

- An uncertainty-quantification study combining MC-Dropout aleatoric/epistemic decomposition, deep-ensemble disagreement, risk–coverage (selective prediction) analysis, and post-hoc temperature scaling.
- A data-audit and ablation phase that rules out duplicate-leakage and label-parsing artefacts, isolates the effect of TF-IDF weighting versus raw bag-of-words counts, and quantifies the gain from validation-tuned decision thresholds.

1.5 Significance of the Study

Beyond its technical contributions, this thesis is intended to be practically useful to three audiences. For platform moderators, the comparative benchmark identifies which model family offers the best accuracy-for-cost trade-off, which matters directly for the compute budget of a moderation pipeline that may need to score millions of posts per day. For researchers working on other low-resource languages, the five-track evaluation harness and its associated statistical-testing methodology (bootstrap confidence intervals, McNemar tests) provide a template that can be reused with a different corpus or language with minimal modification. For the broader discussion of automated content moderation, the explainability and uncertainty-quantification results speak directly to the recurring criticism that machine-learning moderation systems are opaque “black boxes”, by demonstrating concretely what a Bengali hate-speech classifier attends to and how confident it is in each of its own predictions.

1.6 Stakeholders and Intended Use

Three broad classes of stakeholder are relevant to this work. Platform operators (Bengali-language social networks and video platforms) are the most direct beneficiaries, since a validated comparative benchmark tells them which model family to deploy for a given compute budget. Content moderators are a second stakeholder group, whose day-to-day workload could be reduced if a well-calibrated classifier pre-filters obviously non-bullying posts and surfaces uncertain or clearly abusive posts for review, together with an explanation of why a post was flagged. Ordinary platform users, and particularly the minors and public figures who are disproportionately targeted by online harassment, are the ultimate intended beneficiaries, since faster and more accurate detection of bullying content should, all else equal, translate

into a less hostile platform experience. This thesis does not build or evaluate a deployed moderation system; it instead provides the empirical and methodological groundwork model comparison, fusion/ensembling, explainability, and uncertainty quantification that such a system would need.

1.7 Scope and Limitations

This study is anchored on a single public Bengali corpus and a binary bullying/non-bullying label; while the full comparative protocol is additionally replicated end-to-end on a second, independent corpus (BD-SHS; Section 3.9, Section 4.8) to test whether the model ranking generalises beyond one dataset, this study does not evaluate code-mixed Banglish text explicitly or multi-label hate-speech categories.

1.8 Thesis Organization

The remainder of this thesis is organized as follows. Chapter II reviews related work on Bengali and general-purpose hate-speech/cyberbullying detection and social-media forensics. Chapter III describes the dataset, preprocessing pipeline, feature representations, model architectures, fusion/ensemble strategies, and experimental protocol. Chapter IV presents the implementation, comparative results, explainability analysis, uncertainty-quantification study, ablations, and the extent to which the objectives stated in Section 1.3 were achieved. Chapter V presents the concluding part of the thesis, including a summary of the major findings, the limitations of the study, directions for future work, and the concluding remarks.

1.9 Chapter Summary

This chapter has motivated the problem of automated Bengali cyberbullying and hate-speech detection, framed it as a binary classification task complicated by class imbalance, morphological richness, and code-mixed input, stated the five objectives this thesis pursues, and outlined its five main contributions: a controlled multi-track benchmark, a fusion/ensembling

study, a four-view explainability analysis, an uncertainty-quantification study, and a data-audit/ablation phase. The next chapter situates these contributions within the existing literature on Bengali hate speech, sequence and transformer modelling, explainability, and social-media forensics.

Chapter II

Literature Review

This chapter surveys the research landscape that this thesis builds on, organised into six parts: Bengali-specific hate-speech and cyberbullying detection (Section 2.1), the sequence-model and transformer architectures used across the modelling tracks (Section 2.2), general (non-Bengali) hate-speech and cyberbullying detection (Section 2.3), explainability and uncertainty-quantification techniques adopted in this thesis (Section 2.4), the broader social-media-forensics literature that frames the motivation of this work (Section 2.5), and a comparison of the most relevant prior studies (Section 2.6), while Section 2.8 summarises the chapter.

2.1 Bengali Hate-Speech and Cyberbullying Detection

Early Bengali hate-speech research relied on classical machine learning over lexical features. Romim et al. [1] released a 30,000-comment YouTube/Facebook corpus spanning seven categories (sports, entertainment, religion, politics, crime, celebrity, and memes/TikTok) and reported an SVM baseline reaching 87.5% accuracy, with GRU-based deep models evaluated alongside; the corpus used in this thesis is the same public release, which makes their reported numbers a useful external reference point for Chapter 4. Karim et al. [2] proposed a multichannel convolutional-LSTM architecture for under-resourced Bengali classification benchmarks, demonstrating that combining multiple convolutional channels over word- and character-level input improves robustness on small Bengali corpora, an intuition echoed in this thesis’s own character-n-gram TF-IDF channel (Section 3.3). Karim et al. subsequently introduced DeepHateExplainer [3], which paired a transformer ensemble (Bangla-BERT-base, multilingual BERT, and XLM-RoBERTa) with sensitivity analysis and layer-wise relevance propagation to produce human-interpretable explanations for Bengali hate-speech predictions a motivation this thesis shares in Section 4.5, though SHAP and LIME are used here rather

than layer-wise relevance propagation, since they are model-agnostic and can be applied consistently across both the lexical and transformer tracks. Larger benchmarks have since appeared, including the BD-SHS corpus of over 50,000 annotated comments [4] and the multi-category HateCorpBN-XL corpus introduced alongside the HateBertBN model [5]. Rather than treating BD-SHS purely as future work, this thesis re-runs its full comparative protocol on BD-SHS as an independent replication corpus (Section 3.9, Section 4.8).

2.2 Sequence Models and Transformer Encoders

On the modelling side, TextCNN [6] and attention-augmented BiLSTMs built on Bahdanau-style additive attention [7] remain competitive sequence-level baselines once initialised from in-domain static embeddings. TextCNN treats a sentence as a matrix of word vectors and applies parallel convolutional filters of different widths to capture local n-gram-like patterns, which is well suited to the short, punchy phrasing typical of abusive social-media comments. The BiLSTM with additive attention instead processes the sentence sequentially in both directions and learns to weight individual time-steps, which allows the network to concentrate on a small number of decisive tokens.

Transformer encoders BERT [8], its distilled variant DistilBERT [9], RoBERTa [10], and the replaced-token-detection objective of ELECTRA [11] have been adapted to Bengali through BanglaBERT and related Bangla-specific pretraining [12], which is used directly in the transformer track of this thesis. BanglaBERT follows the standard masked-language-modelling pretraining objective of BERT but is pretrained from scratch on a large Bengali web corpus, while BanglaELECTRA instead uses the more sample-efficient replaced-token-detection objective of ELECTRA, in which a small generator network proposes plausible token replacements and a larger discriminator network is trained to detect which tokens were replaced; this objective is reported in the original ELECTRA paper to reach comparable downstream performance with substantially less pretraining compute, which is consistent with BanglaELECTRA’s edge over BanglaBERT observed in Chapter 4. Compared to this literature, the contribution of this thesis is not a new architecture but a controlled comparison across architecture families under one protocol, together with fusion, ensembling, explainability, and uncertainty analyses that are rarely reported jointly for Bengali abusive-language detection.

A methodological point worth noting is that TextCNN, BiLSTM, and transformer encoders make progressively weaker assumptions about which parts of the input matter: TextCNN’s convolutional filters have a fixed receptive field set by the kernel width, so they can only directly capture local n-gram-like patterns; the BiLSTM’s recurrent state can in principle carry information across the whole sequence, but in practice recurrent networks are known to weight nearby tokens more heavily than distant ones; and the transformer’s self-attention mechanism computes a pairwise interaction between every pair of tokens in a single layer, with no distance-based prior at all. This progression from local, to sequential-with-recency-bias, to fully pairwise attention is a plausible partial explanation for the accuracy ordering observed in Table 4.1 (TextCNN and BiLSTM trailing the transformers), though the direct attention comparison is the more direct evidence offered in this thesis for how the two families actually differ in practice on this corpus.

2.3 General (Non-Bengali) Hate-Speech and Cyberbullying Detection

Outside Bengali, hate-speech and cyberbullying detection is a mature NLP sub-field with well-established benchmark shared tasks (e.g., OffensEval/SemEval and the TRAC aggression-identification shared tasks), typically framed as binary or multi-class text classification over social-media posts. Classical approaches combined TF-IDF or bag-of-words features with SVMs or logistic regression; subsequent work moved to word embeddings (Word2Vec, GloVe, FastText) feeding CNN/LSTM classifiers, and more recently to fine-tuned transformer encoders, which now define the state of the art across most high-resource-language hate-speech benchmarks. A recurring finding across this literature echoed in Section 4.5 of this thesis is that lexical/surface cues (profanity, slurs, second-person pejoratives) remain strong predictors even for transformer-era models, while implicit hate speech (coded language, sarcasm, dog-whistles) remains comparatively hard to detect regardless of architecture. This general pattern motivated including a dedicated character-n-gram TF-IDF channel (Section 3.3) and a mechanistic attention comparison in the present study, since both are aimed specifically at the surface-cue-versus-implicit-cue distinction. Social-media forensics research on criminal-behaviour identification (Section 2.5) sits adjacent to this literature: where mainstream hate-speech detection asks “is this post abusive?”, the forensics framing additionally asks “what

does this pattern of posting reveal about the poster, and should it be escalated?” a distinction that motivates treating the bullying/non-bullying classifier developed in this thesis as one input to a larger moderation or escalation workflow rather than as a final automated decision-maker.

2.4 Explainability and Uncertainty Quantification

For interpretability, this thesis draws on SHAP [13] for global additive feature attributions and LIME [14] for local, model-agnostic case explanations. SHAP assigns each feature a Shapley value drawn from cooperative game theory, quantifying its average marginal contribution to a prediction across all possible feature subsets, which gives a theoretically grounded notion of global feature importance. LIME instead perturbs the input around a single instance, fits an interpretable surrogate model (typically sparse linear regression) to the perturbed predictions, and reports the surrogate’s coefficients as a local explanation a technique well suited to inspecting individual confusion-matrix cases. For uncertainty, MC-Dropout is used as a practical Bayesian approximation [15], in which dropout is kept active at inference time and repeated stochastic forward passes approximate samples from an implicit posterior over model weights; this is combined with deep-ensemble disagreement [16], which instead trains several independently initialised models and treats their disagreement as an uncertainty signal, and post-hoc temperature scaling for calibration [17], which rescales the logits of an already-trained network by a single learned scalar temperature to better align predicted confidence with empirical accuracy following standard practice for the trustworthy deployment of deep text classifiers.

2.5 Broader Context: Social Media Forensics and Behaviour Identification

Beyond language-specific hate-speech detection, this work is also motivated by the broader field of social media forensics, which studies how text posted on social platforms can reveal a person’s aggressive, abusive, or offensive behavioural traits and, in extreme cases, support crime prevention by cyber-security and law enforcement agencies. Ashraf et al. [18] proposed a framework for criminal behaviour identification from social media in which posted text is first filtered for relevance and then classified into behavioural categories offensive, aggressive,

frustrated, and abusive grounded in Maslow’s hierarchy of needs, arguing that unmet psychological needs (physiological, safety, love/belonging, esteem, and self-actualisation) are reflected in negative online behaviour, and that identifying such behaviour early can support crime-prevention efforts by law enforcement and cyber-security agencies. Their proposed system architecture separates data relevant to a person’s behaviour from irrelevant data, then further classifies the relevant data into behavioural sub-classes, with profiles exhibiting a criminal nature of behaviour stored in a database accessible to law-enforcement agencies for cross referencing against existing records. Methodologically, Ashraf et al. [18] used a deep multilayer-perceptron classifier over hybridised unigram, bigram, and trigram TF-IDF/bag-of-words/GloVe features on a cyber-troll Twitter dataset of 20,001 tweets, and reported F1 scores of 87% (unigram + bigram) and 88% (bigram + trigram), improving on a reproduced state-of-the-art baseline that had originally been reported at 90% F1 but which they found reproduced at only 84% F1 under their own re-implementation. Their central argument that aggressive online behaviour is worth detecting early because it can precede real-world harm, and that hybridising multiple n-gram/embedding views improves detection over any single view directly parallels the design of this thesis: the same intuition that combining TF-IDF, static embeddings, and contextual representations should outperform any single view motivates the fusion and ensemble tracks described in Section 3.5, and the shared framing of “behaviour detection for prevention” motivates treating explainability (Section 4.5) as a first-class requirement rather than an afterthought, since a system used to flag potential criminal or harmful behaviour must be able to justify its decisions to a human reviewer.

2.6 Related-Work Comparison

Table 2.1 summarises the studies most directly comparable to this thesis across five dimensions: language, model families evaluated, dataset size, best reported F1, and whether explainability and/or uncertainty quantification were reported. The comparison highlights that no prior Bengali study reports all of comparative multi-family benchmarking, statistical significance testing, explainability, and uncertainty quantification together, which is precisely the combination this thesis targets.

Table 2.1: Comparison with Related Studies

Study	Language	Model Families	Best F1	XAI	Uncertainty
Romim et al. [1]	Bengali	SVM, GRU	0.875 (acc.)	No	No
Karim et al. [2]	Bengali	Multichannel CNN-LSTM	0.9045	No	No
Karim et al. [3] (DeepHateExplainer)	Bengali	BERT ensemble (mBERT, XLM-R, BanglaBERT)	0.78– 0.91	Yes (LRP)	No
Ashraf et al. [18]	English (Twitter)	MLP + hybrid TF-IDF/BoW/GloVe	0.87– 0.88	No	No

2.7 Research Gap

Taken together, the literature reviewed above shows :

- That Bengali-specific hate-speech resources and benchmarks are comparatively young and fragmented across corpora of differing size and label granularity.
- That most prior studies, including Ashraf et al.’s social-media-forensics framework, report a single model family rather than a controlled cross-family comparison.
- That explainability is usually reported for a single interpretability method rather than compared across global, local, and mechanistic views.
- That predictive-uncertainty quantification is rarely reported for Bengali abusive-language models despite its importance for real-world moderation deployment

This thesis is designed to close these four gaps simultaneously within one evaluation harness, on the same public Bengali corpus used by the original dataset authors [1].

2.8 Chapter Summary

This chapter reviewed Bengali-specific hate-speech detection literature, the sequence-model and transformer architectures adopted in Chapter 3, general (non-Bengali) hate-speech research, the explainability and uncertainty-quantification techniques used in Chapter 4, and

the social-media-forensics literature that frames this thesis’s motivation, before summarising how the present study’s five contributions close specific gaps in that literature. The next chapter details the dataset, preprocessing, feature representations, model architectures, and experimental protocol used to generate the results reported in Chapter 4.

Chapter III

Methodology

This chapter describes the dataset, preprocessing pipeline, feature representations, the five modelling tracks, the fusion/ensemble strategies, and the experimental protocol used throughout the thesis. All models are registered through a single evaluation harness (implemented in Python/Jupyter) so that every downstream comparison the master results table, bootstrap intervals, McNemar tests, fusion, and ensembling draws on an identical set of stored validation/test probabilities. Figures 3.1, 3.5, and 3.6 together summarise the overall pipeline: raw comments are cleaned into two parallel text views, split 70/15/15, transformed into five families of representation, classified by seventeen registered models, and finally scored by a common eleven-metric evaluation harness.

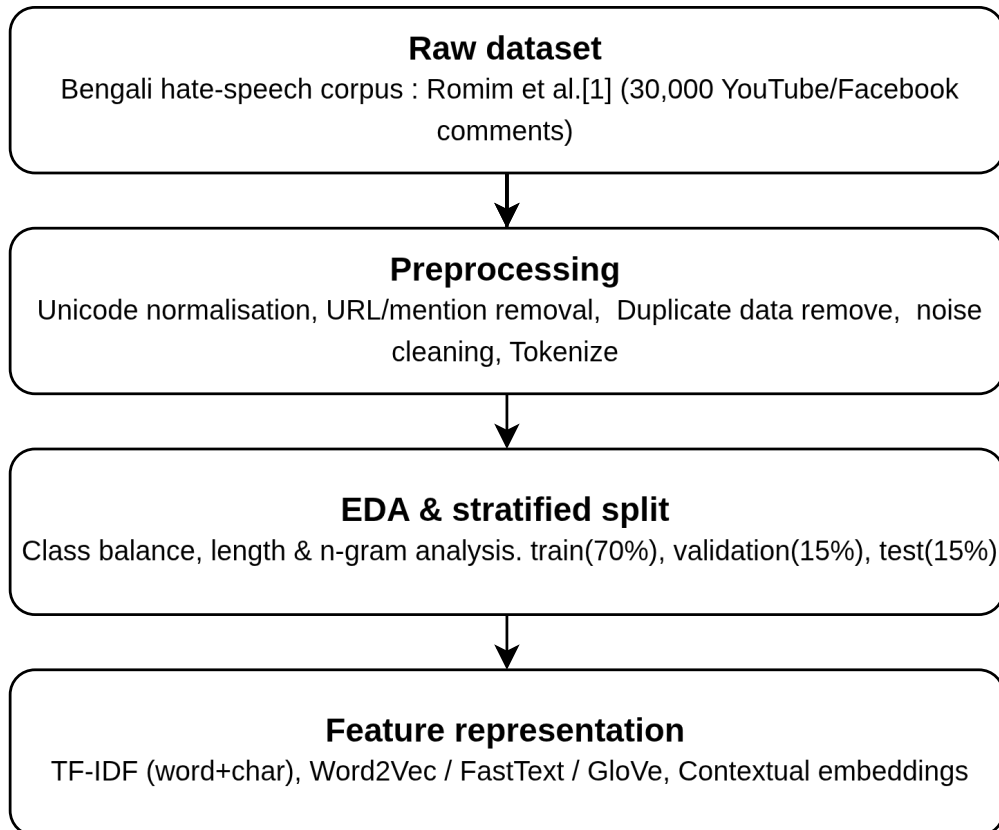


Figure 3.1: Overview of the data preparation pipeline: raw corpus preprocessing EDA & stratified split feature representation.

3.1 Dataset

This thesis uses the public Bengali hate-speech corpus of Romim et al. [1], comprising 30,000 raw comments harvested from YouTube and Facebook and manually labelled by crowd annotators (majority vote of three annotators per comment) as bullying/hate (1) or non-bullying (0), together with one of seven source categories. After dropping empty/NA rows and exact-duplicate text (Section 4.7.1), 29,829 comments remain, with a 33.4% / 66.6% positive/negative split (Table 3.1). Crime-related content carries the highest hate rate (42.0%) and celebrity content the lowest (20.2%), consistent with the topical skew typically reported for this corpus (Table 3.2).

The choice of this particular corpus, rather than collecting a new one, was deliberate: annotating a comparably sized Bengali hate-speech dataset from scratch would require multiple trained annotators, an agreed annotation guideline, and an inter-annotator-agreement study, none of which is feasible within the scope of a single thesis, and reusing a public, previously validated corpus additionally makes the results in Chapter 4 directly comparable to the original dataset authors’ own reported baseline (Section 2.1) and to any future work that uses the same release. The majority-vote annotation scheme (three annotators per comment) used by the original authors is a standard mitigation against single-annotator idiosyncrasy, though it does not eliminate systematic disagreement about borderline cases a point revisited in the discussion of the residual error rate in Section 4.7.1 and the threats-to-validity discussion in Section 4.9.

Table 3.1: Dataset statistics (post-cleaning)

Property	Value
Original dataset	30,000
After dataset cleaning	29,836
Removed during preprocessing	7
Final dataset used	29,829
Bullying (1)	9,958 (33.4%)
Non-bullying (0)	19,871 (66.6%)
Avg. words/post	14.69
Median words/post	9
Vocabulary	62,419

Table 3.2: Category Distribution and Hate Rate

Category	N	Hate %
Crime	4,743	42.0%
Entertainment	4,519	36.0%
Sports	4,392	36.3%
Religion	4,357	36.0%
Memes/General	4,205	36.9%
Politics	3,958	22.3%
Celebrity	3,662	20.2%

3.2 Exploratory Data Analysis

Figure 3.2 shows the overall class balance together with the post-length distribution by class; bullying posts are on average shorter and more variable in length than non-bullying posts, which motivates the inclusion of a character-level TF-IDF channel (Section 3.3) that remains informative even on very short inputs.

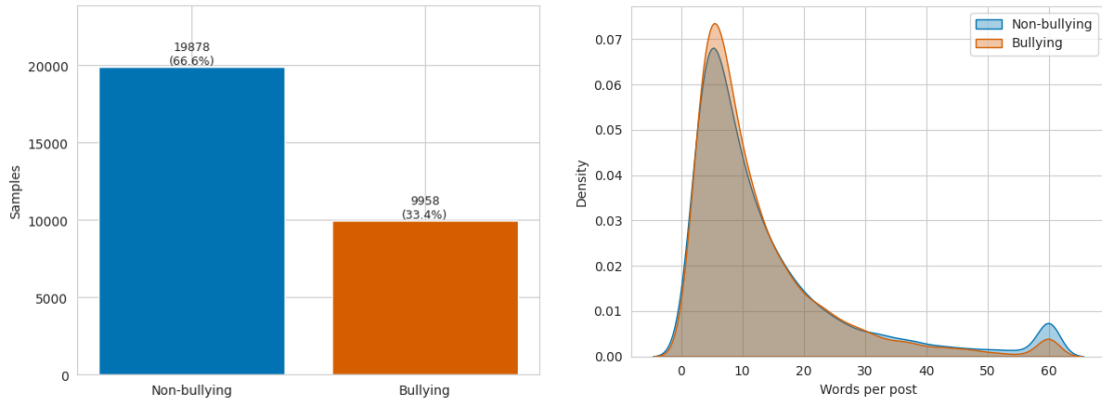


Figure 3.2: Class balance (left) and post-length distribution by class (right).

Figure 3.3 contrasts the most frequent unigrams for bullying versus non-bullying posts after stop-word-preserving Bangla-aware tokenisation; bullying posts are dominated by profanity and second-person invective, while non-bullying posts are dominated by topical and neutral vocabulary. Word clouds (Figure 3.4) corroborate this split visually.

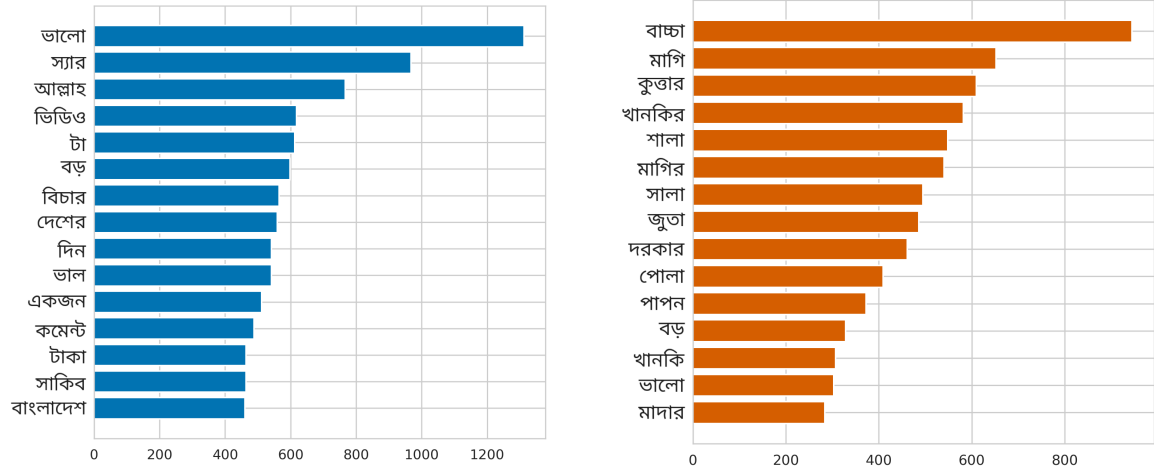


Figure 3.3: Top unigrams for non-hate speech (left) and hate speech (right) posts, by frequency.



Figure 3.4: Word clouds for non-hate speech (left) and hate speech (right) posts, rendered with a Bangla-capable font.

3.3 Text Cleaning, Feature Representations, and Splits

Two parallel text views are produced. The clean view lower-cases text, replaces URLs and @-mentions with placeholder tokens, and retains the Bengali Unicode block (U+0980–U+09FF), Bengali sentence punctuation, and Latin letters/digits for code-mixed Banglish tokens, discarding emoji and stray symbols; this view feeds the lexical, static-embedding, and deep-learning tracks. The raw view is left untouched for the transformer tokenisers, which handle noisy text natively via subword segmentation. Data are split 70/15/15 (train/validation/test) with stratification on the label, using a fixed random seed; the positive rate is preserved at 33.4% in every split (Table 3.3). All model-selection decisions early stopping, fusion weights, stacking coefficients, temperature scaling use the validation split only, and the test split is touched exactly once per model for the final metrics reported in Chapter 4.

Table 3.3: Train/Validation/Test split statistics

Split	N (share)	Bullying %
Train	20,879 (70%)	33.4%
Validation	4,475 (15%)	33.4%
Test	4,475 (15%)	33.4%

The five feature-representation families used across the modelling tracks are as follows:

- Lexical: word-level TF-IDF (1–2 grams, 30,000 features) concatenated with character-level TF-IDF (3–5 grams, 30,000 features, word-boundary analyzer) to give a 60,000-dimension sparse representation; the character channel is included specifically because it is robust to obfuscated profanity that word-level tokenisation misses.
- Static embeddings: Word2Vec and FastText (skip gram, 100-d, trained on the training split only) and, where available, 100-d GloVe vectors, mean-pooled per post.
- Contextual embeddings: 768 dimensional mean-pooled last hidden state representations extracted from the best fine-tuned transformer, used only in the fusion track.

The TF-IDF weight of term t in document d , drawn from corpus D , is computed as the product of term frequency and inverse document frequency:

$$TF\text{-}IDF(t, d) = tf(t, d) \times \log \left(\frac{|D|}{1 + df(t)} \right) \quad (3.1)$$

where $tf(t, d)$ is the raw count of term t in document d , $|D|$ is the total number of documents, and $df(t)$ is the number of documents containing t . This down-weights terms that are common across the whole corpus (which carry little discriminative signal) relative to terms that are concentrated in a smaller subset of documents (which are more likely to be informative for classification).

3.4 Modelling Tracks

3.4.1 Track A Classical Lexical and Static-Embedding Baselines

Logistic regression, linear SVM (Platt calibrated), multinomial Naive Bayes, and XGBoost are trained on the combined TF-IDF representation with class-balanced weighting (linear

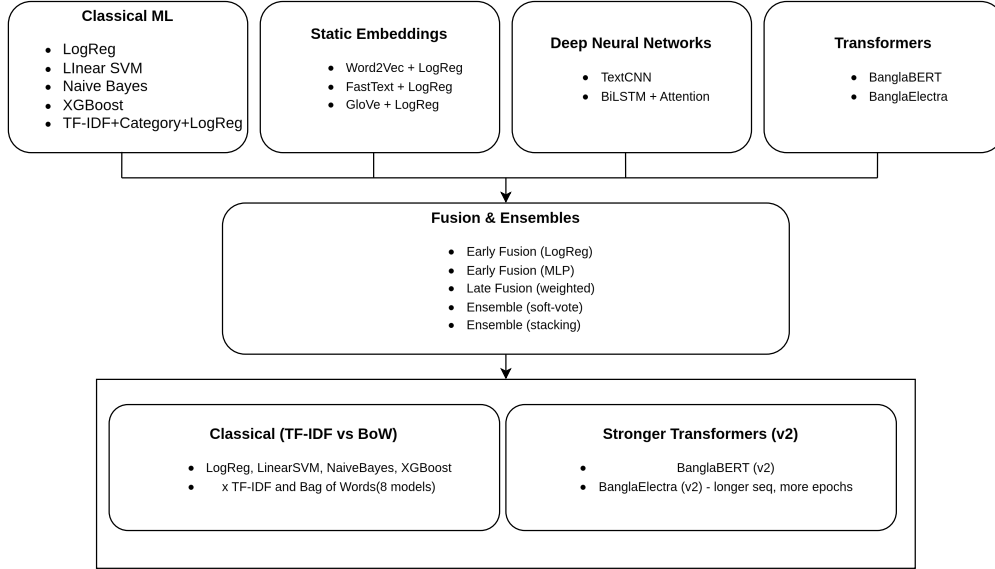


Figure 3.5: Overview of the five modelling tracks, fusion/ensemble strategies, and add-on extended experiments.

models) or a positive-class scale weight (XGBoost). An auxiliary-feature ablation concatenates a one-hot post-category vector onto the TF-IDF space to test whether topical metadata helps a lexical classifier. The mean-pooled static embeddings (Word2Vec, FastText, GloVe) are each fed to a standardised logistic regression for comparison. Logistic regression estimates the probability of the positive class as a sigmoid function of a linear combination of input features:

$$P(y = 1 \mid x) = \frac{1}{1 + \exp(-(w^T x + b))} \quad (3.2)$$

where w is the learned weight vector, b is a bias term, and x is the feature vector (TF-IDF or mean-pooled embedding). This closed-form probabilistic interpretation is what makes the model's coefficients directly usable for the global explainability analysis.

Linear SVM instead seeks the maximum-margin hyperplane separating the two classes, solving (in its soft-margin primal form):

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i, \quad \text{s.t. } y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (3.3)$$

where C controls the trade-off between margin width and misclassification penalty on the slack variables ξ_i ; class-balanced weighting scales C per class in proportion to inverse class

frequency. Because SVM decision scores are not directly probabilities, Platt scaling (a logistic regression fitted on the SVM’s decision-function outputs) is used to obtain calibrated class probabilities for the evaluation harness. Multinomial Naive Bayes instead makes a conditional independence assumption over features given the class, and predicts using Bayes’ rule:

$$P(y | x) \propto P(y) \prod_j P(x_j | y) \quad (3.4)$$

where $P(x_j|y)$ is estimated from the relative frequency of feature j (a TF-IDF/BoW term) within each class in the training data, with Laplace smoothing to avoid zero probabilities for unseen terms. XGBoost is a gradient boosted ensemble of shallow decision trees, in which each new tree is fit to the negative gradient of the loss with respect to the current ensemble’s predictions, regularised by an explicit penalty on tree complexity; a positive class scale weight rebalances the gradient contribution of the minority (bullying) class in proportion to the training set imbalance, analogous to the class balanced weighting used by the linear models.

3.4.2 Track B Deep Neural Networks

TextCNN [6] applies parallel 1-D convolutions (kernel widths 3/4/5, 128 filters each) over token embeddings initialised from the in domain Word2Vec vectors (100% vocabulary coverage), followed by max pooling and a dropout-regularised classification head. A BiLSTM with additive (Bahdanau style) attention [7] processes the same embedding input; the attention distribution is retained for the mechanistic explainability comparison. Given encoder hidden states $h_1 \dots h_T$, additive attention computes a scalar alignment score for each time-step, normalises the scores with a softmax, and forms a weighted context vector:

$$e_t = v^T \tanh(W h_t + b); \quad \alpha_t = \text{softmax}(e_t); \quad c = \sum_t \alpha_t h_t \quad (3.5)$$

where v , W , and b are learned parameters. Both networks are trained with class-weighted cross entropy and early stopping on validation F1-macro. The class-weighted binary cross entropy loss for a mini-batch of N examples is:

$$L = -\frac{1}{N} \sum_i [w_{y_i} (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i))] \quad (3.6)$$

3.4.3 Track C Fine-Tuned Transformers

BanglaBERT and BanglaELECTRA (both from the csebuetnlp Bangla pretraining releases, built on the BERT [8] and ELECTRA [11] objectives respectively) are fine tuned end to end on raw text with mixed-precision (fp16) training, linear learning-rate warm-up, gradient clipping, and class weighted cross entropy loss; the validation best checkpoint is restored before test evaluation. Each input post is tokenised into subword units, prefixed with a special classification token ([CLS]), and passed through the pretrained transformer encoder; the final hidden state of the [CLS] token is fed to a lightweight classification head.

3.4.4 Track D Representation Fusion

Early (feature -level) fusion concatenates a 300-dimensional TruncatedSVD projection of the TF-IDF space, the 100-dimensional Word2Vec mean-pooled vector, and the 768-dimensional contextual transformer embedding into a single 1,168-dimensional joint space, classified with either logistic regression or a single-hidden-layer (256-unit) MLP. Late (decision-level) fusion instead averages the test-set class probabilities of six heterogeneous members: TF-IDF+LogReg, TF-IDF+XGBoost, TextCNN, BiLSTM+Attention, BanglaBERT, and BanglaELECTRA, with weights proportional to each member’s validation F1-macro:

$$P_{\text{fused}}(y = 1 \mid x) = \sum_m w_m P_m(y = 1 \mid x), \quad w_m \propto F1\text{-macro}_m(\text{validation}) \quad (3.7)$$

3.4.5 Track E Ensembling

Soft voting averages the probabilities of the four strongest members by validation F1-macro. Stacked generalisation trains a logistic regression meta-learner on the six members’ validation-set probabilities (the standard held-out stacking protocol, avoiding information leakage) and applies the fitted meta-learner once to the corresponding test-set probabilities. Held-out stacking is deliberately used instead of in-sample stacking because fitting the meta-learner on the

same predictions used to train the base models would leak information and produce an overly optimistic estimate of the ensemble’s generalisation performance.

3.5 Evaluation Protocol

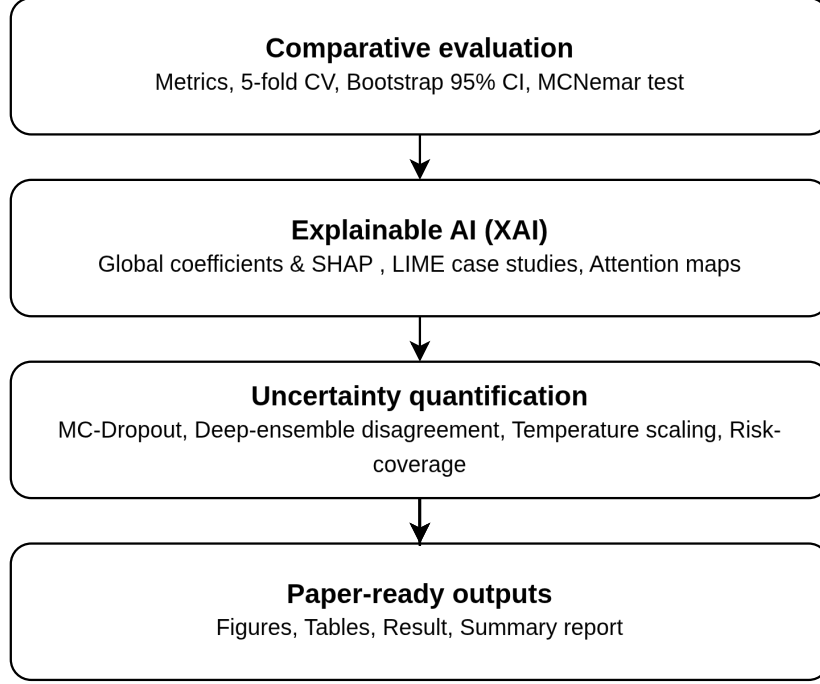


Figure 3.6: Overview of the evaluation, explainability, and uncertainty-quantification pipeline.

Every model is scored on the held-out test split ($n = 4,475$) with eleven metrics: accuracy, precision, recall, F1 (positive class), F1-macro, ROC-AUC, PR-AUC, Matthews correlation coefficient (MCC), Brier score, and Expected/Maximum Calibration Error (ECE/MCE, 10 bins), plus fit time, per-sample inference latency, and parameter count where applicable. F1-macro is used as the primary model selection criterion because it weights both classes equally under the 1:2 class imbalance:

$$F1\text{-macro} = \frac{F1_{\text{bullying}} + F1_{\text{non-bullying}}}{2} \quad (3.8)$$

A 5-fold cross-validation protocol is additionally run for the four classical models, re-fitting the TF-IDF vectoriser inside every fold to avoid information leakage. Paired bootstrap re-sampling (1,000 resamples of the test indices) yields 95% confidence intervals for F1-macro, and continuity-corrected McNemar tests give pairwise significance between the top models’

error patterns, using the standard McNemar test statistic computed from the discordant-pair counts b and c (cases where the two compared models disagree):

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (3.9)$$

3.6 Hardware and Software

Experiments were run on a Kaggle NVIDIA T4 GPU accelerator with CUDA-enabled PyTorch, using HuggingFace Transformers for tokenisation and fine-tuning, scikit-learn for classical models and metrics, gensim for Word2Vec/FastText, XGBoost for gradient boosting, and SHAP/LIME for explainability. Total wall-clock runtime for the full pipeline (all tracks, all figures) was approximately 3 hours.

Table 3.4: Key Hyperparameters

Component	Key settings
TextCNN	128 filters $\times$ {3,4,5}, dropout 0.5, Adam, class-weighted CE, early stop on val F1-macro
BiLSTM + Attention	Bidirectional, additive attention, dropout 0.5, Adam, early stop on val F1-macro
BanglaBERT / BanglaELECTRA	max_len 64 (128 in extended re-run), batch 32, linear warm-up, fp16, class-weighted CE, 3 epochs (5 in extended re-run) with checkpoint restore
Early Fusion (MLP)	1 hidden layer (256 units), early stopping, Standard-Scaler input
Stacking meta-learner	Logistic regression on 6-member validation probabilities

3.7 Explainability and Uncertainty-Quantification Methods

Four complementary explainability views are produced: (i) global feature importance via logistic-regression coefficients and SHAP [13] over the TF-IDF space; (ii) local, model-agnostic case explanations via LIME [14] on representative examples from each confusion-matrix quadrant; (iii) a mechanistic comparison of transformer CLS-attention against BiLSTM additive-attention weights on the same test post. Predictive uncertainty is quantified

along four axes: MC-Dropout [15] aleatoric/epistemic decomposition, deep-ensemble disagreement (the variance in predicted probability across several independently trained models), risk-coverage (selective prediction) analysis, and post-hoc temperature scaling [17] for calibration, fitted on the validation logits. The predictive entropy for a Bernoulli prediction with probability p is:

$$H(p) = -[p \log(p) + (1 - p) \log(1 - p)] \quad (3.10)$$

and Expected Calibration Error over B equal-frequency confidence bins is computed as:

$$ECE = \sum_{b=1}^B \frac{n_b}{N} \times |acc(b) - conf(b)| \quad (3.11)$$

where n_b is the number of predictions in bin b , N is the total number of predictions, and $acc(b)$ and $conf(b)$ are the empirical accuracy and mean predicted confidence within that bin, respectively.

3.8 Data Audit and Ablation Design

To confirm that the observed performance ceiling reflects genuine task difficulty rather than a pipeline defect, the exact loading and cleaning funnel is reproduced with per-stage class counts. Three ablations are additionally run:

- TF-IDF weighting versus raw bag-of-words counts, holding the n-gram range fixed.
- Validation-tuned decision thresholds versus the default 0.5 threshold.
- Concatenating a one-hot post-category feature onto the TF-IDF space.
- As a robustness check, a stronger transformer configuration with longer context and more training epochs.

3.9 Cross Corpus Replication Protocol (BD-SHS)

The single corpus limitation flagged in Section 1.7 and revisited in Section 5.3 is addressed directly here rather than left purely as future work. To test whether the comparative ranking established in Chapter 4 is specific to the Romim et al. corpus [1] or generalises to an independently collected and annotated Bengali hate-speech corpus, the complete evaluation protocol of Sections 3.4–3.5 is re-run end-to-end on BD-SHS [4], a second, publicly available Bengali hate-speech corpus. BD-SHS is obtained from its released train/validation/test files and merged into a single pool of 50,281 comments before re-applying the identical cleaning pipeline described in Section 3.3 (Unicode normalisation, URL/mention placeholder substitution, duplicate removal, and empty-string filtering). After cleaning, 50,264 comments remain: 24,149 labelled hate speech (48.0%) and 26,115 non-hate speech (52.0%) a near-balanced label distribution that contrasts with the primary corpus’s 33.4%/66.6% split (Table 3.1). The cleaned pool is re-split 70/15/15 with label stratification and a fixed seed, yielding 35,184/7,540/7,540 train/validation/test comments (Table 3.5).

Sixteen of the seventeen main-corpus architectures are re-trained and re-evaluated from scratch on BD-SHS’s own training split, all Tracks A–E models except the auxiliary category- augmented variant, since BD-SHS’s public release does not carry the seven-category source metadata that variant depends on. This is a full replication, not a transfer-learning evaluation: no weights are carried over from the primary-corpus models, and BD-SHS’s own training split is used throughout. The same eleven-metric evaluation harness, paired-bootstrap 95% confidence intervals, and continuity corrected McNemar significance testing described in Section 3.5 are applied unchanged, so that the two corpora’s results are directly comparable in protocol even where their label balance differs.

Two questions are asked of the resulting BD-SHS results in Section 4.8: first, whether the same model (or model family) wins on both corpora in absolute terms; and second the more statistically meaningful question given the two corpora’s differing label balance and size whether the sixteen shared models are ranked in the same relative order on both corpora, measured with Spearman rank correlation between each model’s F1-macro rank on the Romim corpus and its F1-macro rank on BD-SHS.

Table 3.5: BD-SHS corpus statistics and stratified split.

Property	Value
Total comments (raw release)	50,281
After cleaning	50,264
Hate speech (label = 1)	24,149 (48.0%)
Non-hate speech (label = 0)	26,115 (52.0%)
Train / Validation / Test split	35,184 / 7,540 / 7,540

3.10 Chapter Summary

This chapter described the 29,829-post Bengali hate-speech corpus and its 70/15/15 stratified split, the two parallel cleaning pipelines feeding the lexical/static-embedding/deep-learning tracks and the transformer track respectively, the five feature-representation families, the five modelling tracks (classical, static-embedding, deep-learning, transformer, and fusion/ensemble), the eleven-metric evaluation protocol with bootstrap confidence intervals and McNemar significance testing, and the explainability, uncertainty-quantification, and data-audit/ablation methods used to interrogate the trained models. The next chapter presents the results produced by this methodology.

Chapter IV

Implementation, Results and Discussion

This chapter reports the comparative results of all seventeen registered models, followed by the explainability analysis, the uncertainty quantification study, the data audit and ablations, and a discussion of the findings.

4.1 Master Comparative Results

Table 4.1 and 4.2 reports all metrics for every registered model, sorted by test F1-macro. The stacked ensemble and the soft-voting ensemble are statistically indistinguishable from each other (F1-macro = 0.8951 in both cases, differing only at the fourth decimal place) and both edge out weighted late fusion (0.8944) and the best single transformer, BanglaELECTRA (0.8927).

Table 4.1: Master Results Table A (Test Set, $n = 4,475$), Core Metrics

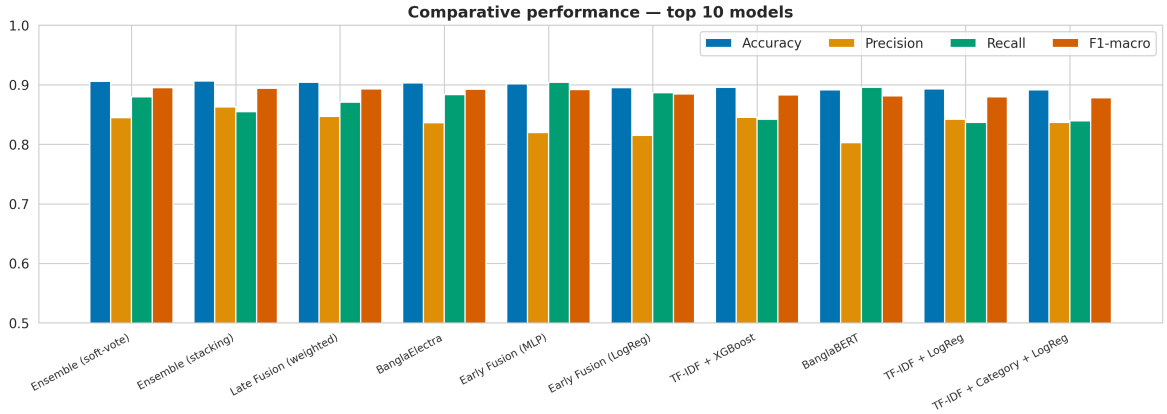
Model	Family	Acc	Prec	Rec	F1	F1-mac
Ensemble (stacking)	Ensemble	0.9068	0.8626	0.8574	0.8600	0.8951
Ensemble (soft-vote)	Ensemble	0.9057	0.8445	0.8795	0.8616	0.8951
Late Fusion (weighted)	Fusion	0.9055	0.8498	0.8708	0.8602	0.8944
BanglaELECTRA	Transformer	0.9032	0.8360	0.8835	0.8591	0.8927
Early Fusion (MLP)	Fusion	0.9032	0.8502	0.8621	0.8561	0.8916
Early Fusion (LogReg)	Fusion	0.8954	0.8143	0.8896	0.8503	0.8850
TF-IDF + XGBoost	Classical	0.8959	0.8454	0.8420	0.8437	0.8828
BanglaBERT	Transformer	0.8916	0.8026	0.8956	0.8466	0.8814
TF-IDF + LogReg	Classical	0.8930	0.8418	0.8367	0.8392	0.8795
TF-IDF + Category + LogReg	Classical	0.8916	0.8366	0.8394	0.8380	0.8783
TF-IDF + LinearSVM	Classical	0.8896	0.8771	0.7784	0.8248	0.8721
TextCNN	Deep NN	0.8769	0.8404	0.7791	0.8086	0.8589
BiLSTM + Attention	Deep NN	0.8691	0.7881	0.8313	0.8091	0.8547
TF-IDF + NaiveBayes	Classical	0.8530	0.7659	0.8059	0.7854	0.8368
FastText + LogReg	Static Emb.	0.8366	0.7401	0.7871	0.7629	0.8191
Word2Vec + LogReg	Static Emb.	0.8192	0.7129	0.7677	0.7393	0.8005
GloVe + LogReg	Static Emb.	0.7140	0.5583	0.6854	0.6154	0.6939

Figure 4.1 compares accuracy, precision, recall, and F1-macro across the top-10 models; the ensembles and fusion models dominate on F1-macro while the calibrated linear SVM

Table 4.2: Master Results Table B (Test Set, $n = 4,475$), Ranking and Calibration Metrics

Model	Family	ROC-AUC	PR-AUC	MCC	ECE
Ensemble (stacking)	Ensemble	0.9639	0.9304	0.7902	0.0258
Ensemble (soft-vote)	Ensemble	0.9637	0.9310	0.7905	0.0231
Late Fusion (weighted)	Fusion	0.9621	0.9287	0.7889	0.0290
BanglaELECTRA	Transformer	0.9605	0.9227	0.7862	0.0324
Early Fusion (MLP)	Fusion	0.9587	0.9273	0.7833	0.0340
Early Fusion (LogReg)	Fusion	0.9541	0.9138	0.7719	0.0421
TF-IDF + XGBoost	Classical	0.9495	0.9170	0.7656	0.0186
BanglaBERT	Transformer	0.9572	0.9218	0.7659	0.0646
TF-IDF + LogReg	Classical	0.9502	0.9080	0.7590	0.0535
TF-IDF + Category + LogReg	Classical	0.9506	0.9099	0.7565	0.0518
TF-IDF + LinearSVM	Classical	0.9455	0.9037	0.7475	0.0178
TextCNN	Deep NN	0.9361	0.8944	0.7192	0.0122
BiLSTM + Attention	Deep NN	0.9352	0.8963	0.7102	0.0321
TF-IDF + NaiveBayes	Classical	0.9202	0.8718	0.6742	0.0748
FastText + LogReg	Static Emb.	0.9042	0.8216	0.6392	0.0464
Word2Vec + LogReg	Static Emb.	0.8831	0.7946	0.6022	0.0488
GloVe + LogReg	Static Emb.	0.7620	0.5957	0.3967	0.0539

attains the single highest precision (0.877) at a lower recall (0.778), reflecting its conservative decision boundary.

**Figure 4.1:** Accuracy, precision, recall, and F1-macro for the top-10 models.

4.2 Statistical Significance

Table 4.3 reports paired-bootstrap 95% confidence intervals for the top ten models: the ensembles' intervals overlap substantially with late fusion and BanglaELECTRA, indicating

the improvement from fusion/ensembling over the best single transformer, while consistent in direction, is modest in absolute size on this dataset and split.

Table 4.3: Master results table (test set, $n = 4,475$), sorted by F1-macro, with 95% bootstrap CIs

Model	F1-mac	95% CI
Ensemble (stacking)	0.8951	[.8869, .9060]
Ensemble (soft-vote)	0.8951	[.8852, .9048]
Late Fusion (weighted)	0.8944	[.8842, .9038]
BanglaELECTRA	0.8927	[.8824, .9025]
Early Fusion (MLP)	0.8916	[.8826, .9019]
Early Fusion (LogReg)	0.8850	[.8741, .8934]
TF-IDF + XGBoost	0.8828	[.8726, .8930]
BanglaBERT	0.8814	[.8718, .8915]
TF-IDF + LogReg	0.8795	[.8699, .8895]
TF-IDF + Category + LogReg	0.8783	[.8685, .8889]
TF-IDF + LinearSVM	0.8721	[.8616, .8829]
TextCNN	0.8589	[.8453, .8679]
BiLSTM + Attention	0.8547	[.8442, .8665]
TF-IDF + NaiveBayes	0.8368	[.8251, .8481]
FastText + LogReg	0.8191	[.8095, .8330]
Word2Vec + LogReg	0.8005	[.7819, .8063]
GloVe + LogReg	0.6939	[.6801, .7069]

Pairwise McNemar significance tests among the top-6 models confirm the three ensemble/fusion variants are not significantly different from one another ($p > 0.05$), while all three are significantly different from the strongest single classical model, TF-IDF+XGBoost ($p < 0.05$) evidence that fusion and ensembling deliver a genuine, if incremental, improvement over the best individually trained model rather than noise.

4.3 Cross-Validation Check

To check that the single split results are not an artefact of a favourable train/validation/test partition, the four classical models are additionally evaluated under 5-fold stratified cross-validation on the combined train+validation data. Cross-validated F1-macro (0.813–0.844) sits within 2–4 points of the corresponding single-split test F1-macro in Table 4.1&4.2, a modest, expected gap consistent with normal sampling variance rather than an unstable or leaky evaluation protocol.

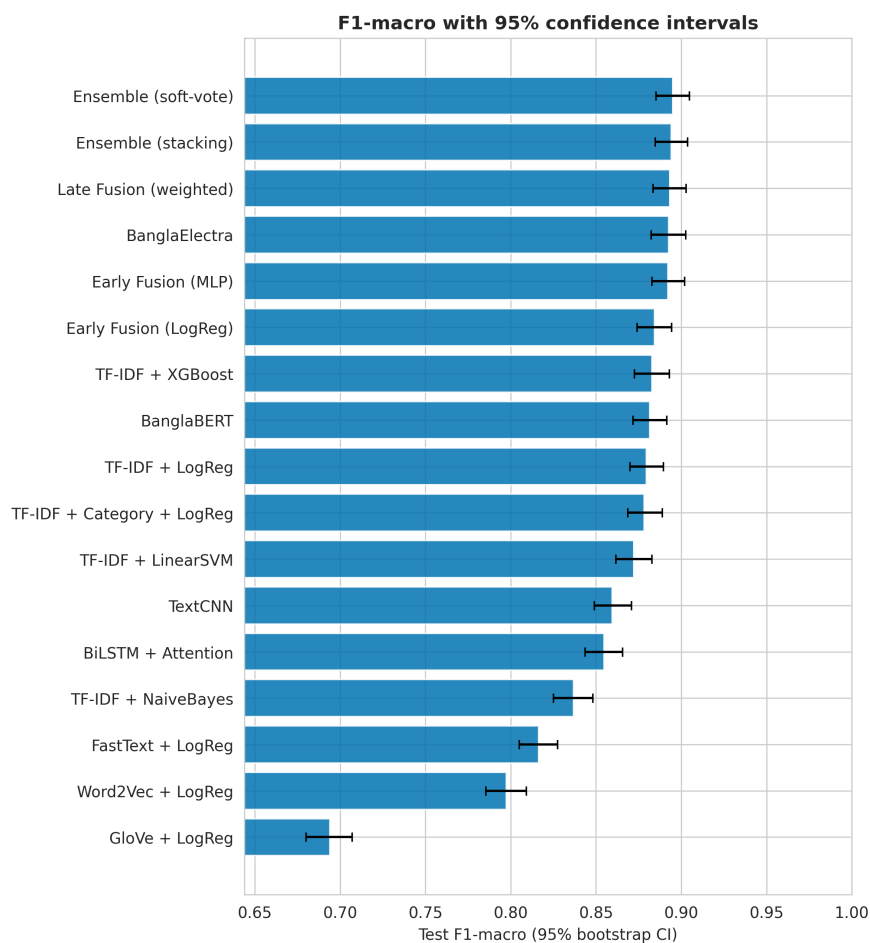


Figure 4.2: F1-macro with bootstrap 95% confidence intervals, all registered models.

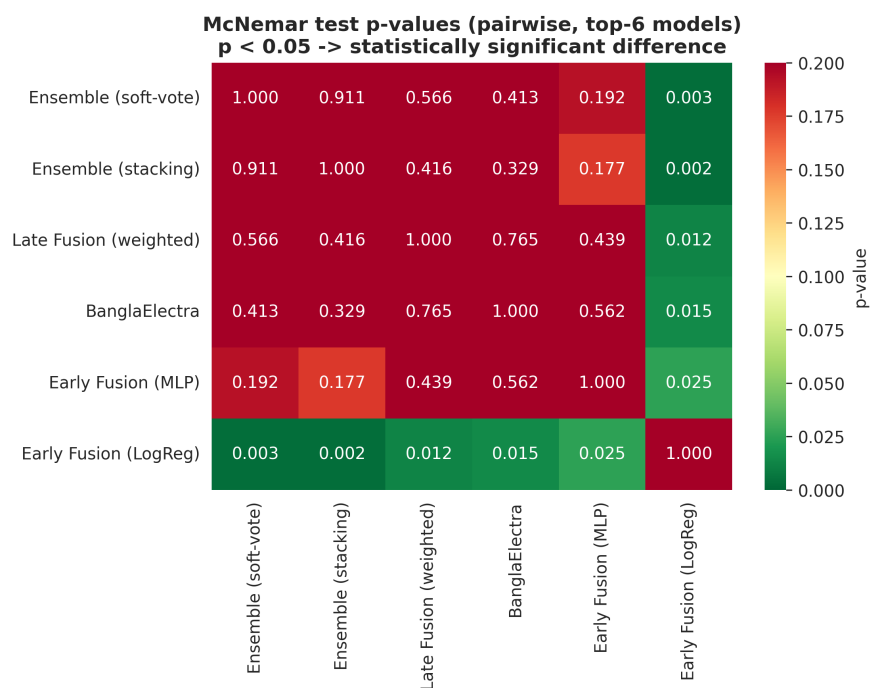


Figure 4.3: Pairwise McNemar significance heatmap, top-6 models by F1-macro.

Table 4.4: 5-Fold Stratified Cross-Validation (Classical Models)

Model	CV F1-macro	CV Accuracy
TF-IDF + LogReg	0.8443 ± 0.0039	0.8616 ± 0.0037
TF-IDF + LinearSVM	0.8432 ± 0.0031	0.8651 ± 0.0027
TF-IDF + NaiveBayes	0.8134 ± 0.0049	0.8424 ± 0.0039
TF-IDF + XGBoost	0.8433 ± 0.0039	0.8646 ± 0.0032

4.4 Training Curves, Embedding Space, and Error Analysis

Figure 4.4 plots training loss and validation F1-macro across epochs for the neural and transformer models, showing BanglaELECTRA converging to the highest validation F1-macro (0.900) within three epochs before mild overfitting sets in, motivating the checkpoint-restore policy described. Figure 4.5 visualises the 768-d contextual embedding space (dimensionality reduced) of the best transformer, with the two classes forming overlapping but separable clusters, consistent with the observed recall/precision trade-off at the default 0.5 threshold.

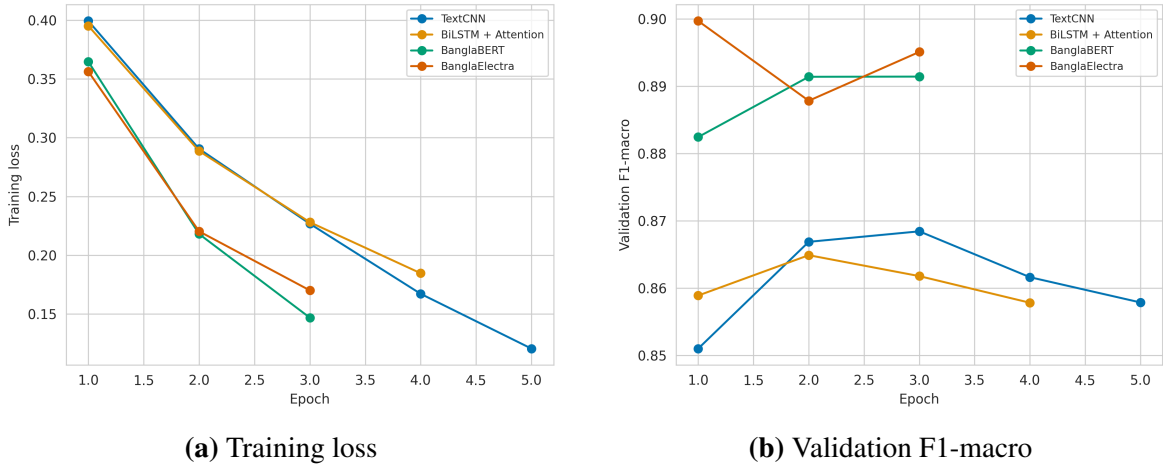


Figure 4.4: Training loss (left) and validation F1-macro (right) across epochs for all neural/transformer models.

ROC and precision-recall overlays (Figure 4.6) show all top 8 models comfortably separating the classes ($\text{ROC-AUC} \geq 0.88$), with the ensembles achieving the highest PR-AUC (≈ 0.93), the more informative curve under 1:2 class imbalance.

The confusion matrix for the best model (Figure 4.7) shows the stacked ensemble correctly identifies 93.2% of non-hate speech posts and 85.7% of hate speech posts, with a false-negative rate (14.3%) that is the primary remaining error mode, consistent with abusive language detection literature, where implicit or sarcastic bullying is harder to recall than lexically

UMAP projection of BanglaElectra embeddings (test set)

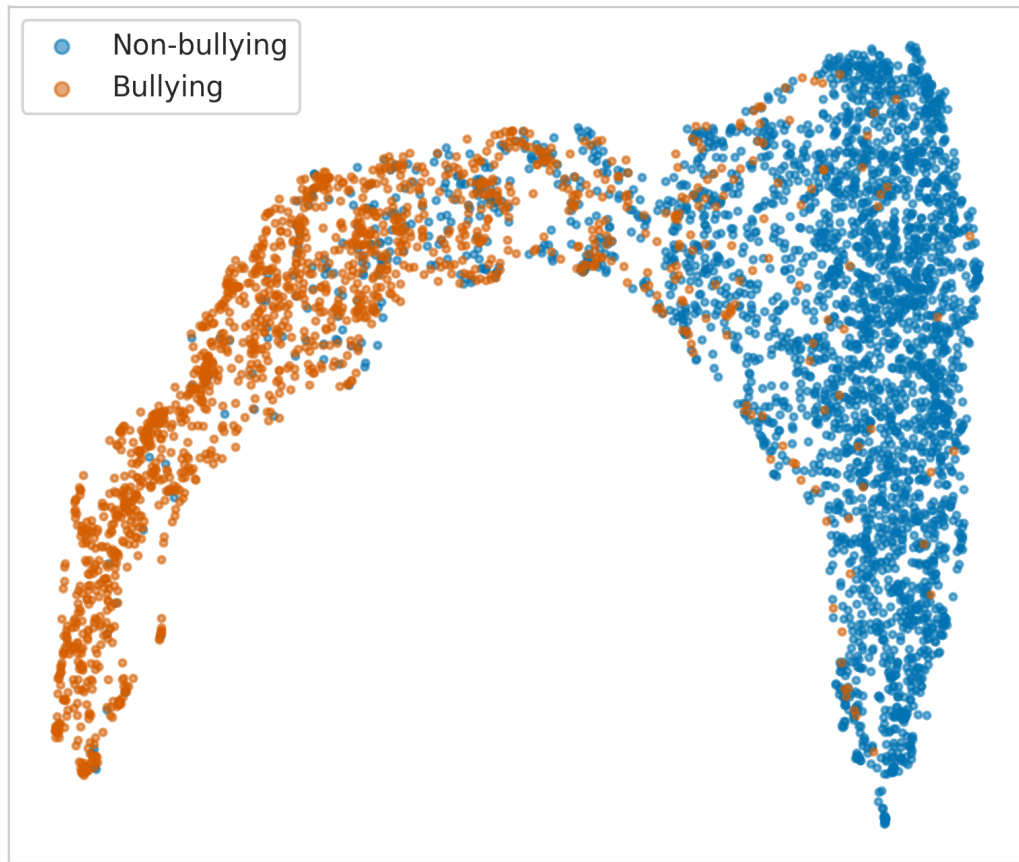


Figure 4.5: Two-dimensional projection of the best transformer’s contextual embedding space, coloured by ground-truth label.

explicit cases.

The multi-metric radar (Figure 4.8) visualises the top-5 models across six axes simultaneously, and the accuracy–efficiency trade-off plot (Figure 4.9) shows that the classical TF-IDF models and the neural networks sit far to the efficiency-favourable side (sub-millisecond to 30 ms per-sample inference) while the fine-tuned transformers require 1 ms per sample on GPU but 110–165M parameters, a relevant consideration for on-device or high throughput moderation pipelines.

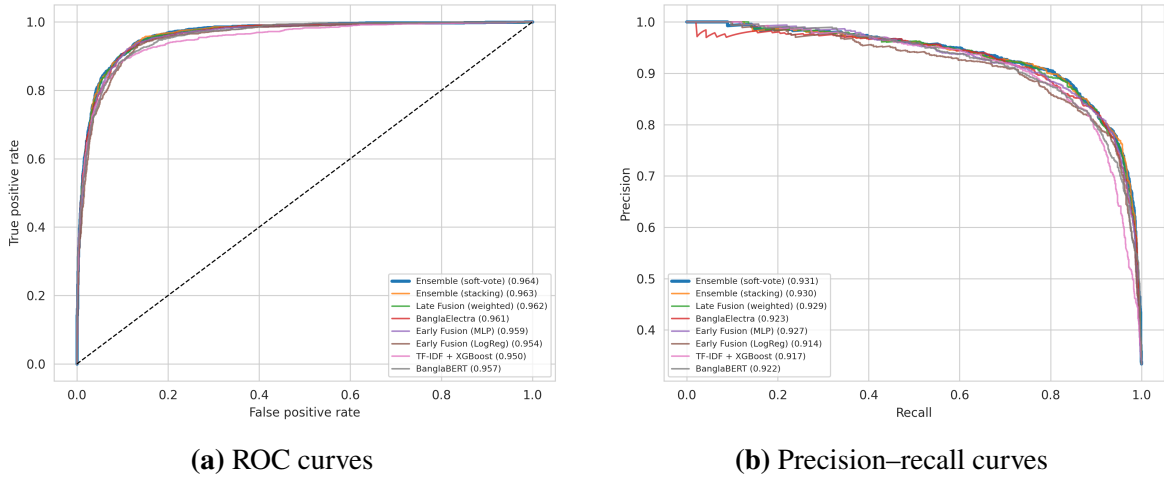


Figure 4.6: ROC curves (left) and precision–recall curves (right) for the top-8 models.

4.5 Explainability Analysis

4.5.1 Global Explanations

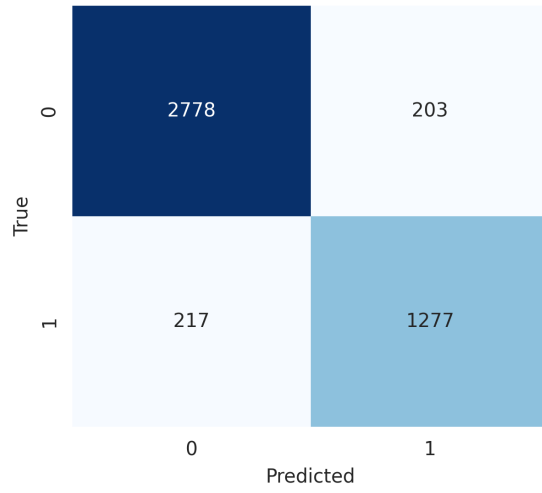
Logistic-regression coefficients over the combined word/character TF-IDF space (Figure 4.10) and a SHAP beeswarm plot over the same space (Figure 4.11) both surface explicit profanity and second-person pejoratives as the dominant positive class drivers, with the character n-gram channel specifically contributing sub-word fragments that catch obfuscated spellings not captured by whole-word features.

4.5.2 Local Explanations (LIME)

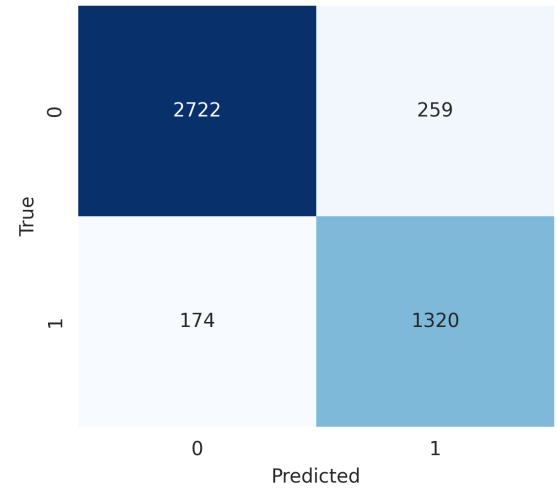
LIME [14] case studies on the best transformer are shown for one representative example from each confusion-matrix quadrant true positive, true negative, false positive, and false negative illustrating which tokens the model weighs most heavily for individual predictions, including cases where sarcasm or indirect phrasing misleads the model (the false-negative case) and where topical but non-abusive vocabulary is over-weighted (the false-positive case).

4.5.3 Mechanistic Comparison: Transformer vs. Recurrent Attention

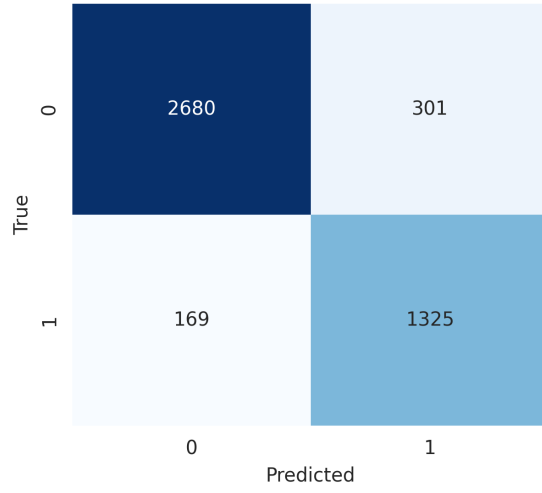
Figure 4.14 compares CLS-attention from the best transformer against the additive-attention weights of the BiLSTM on the same held-out post, revealing that the transformer distributes attention across a broader token span while the BiLSTM concentrates more sharply on one



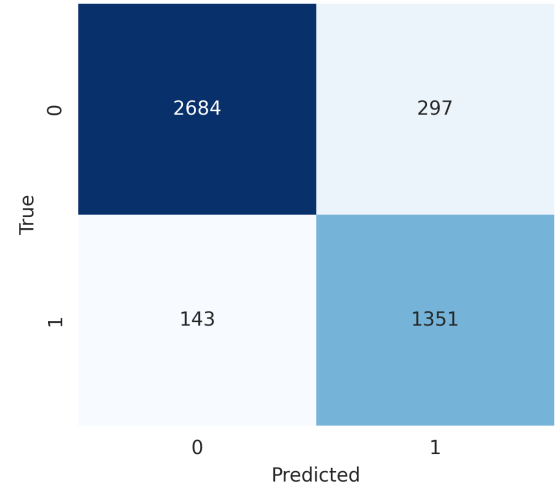
(a) Ensemble (Stacking), F1 = 0.894



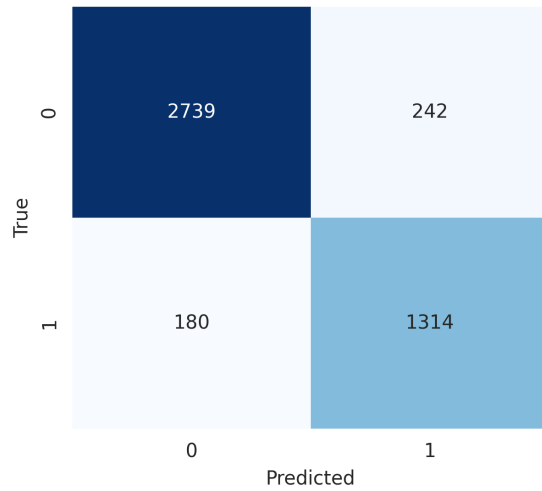
(b) BanglaElectra, F1 = 0.893



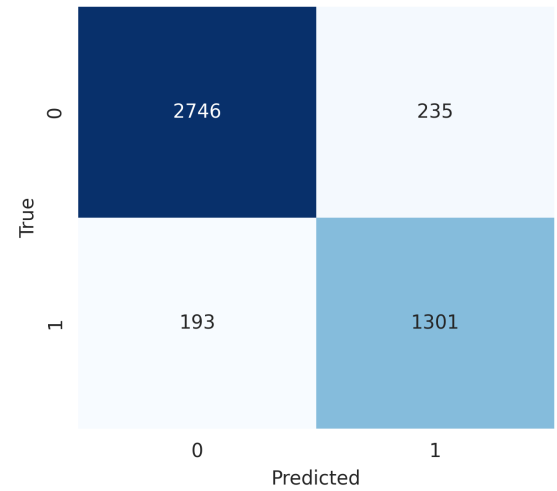
(c) Early Fusion (LogReg), F1 = 0.884



(d) Early Fusion (MLP), F1 = 0.892



(e) Ensemble (Soft Vote), F1 = 0.895



(f) Late Fusion (Weighted), F1 = 0.893

Figure 4.7: Confusion matrices of the top 6 performing models.

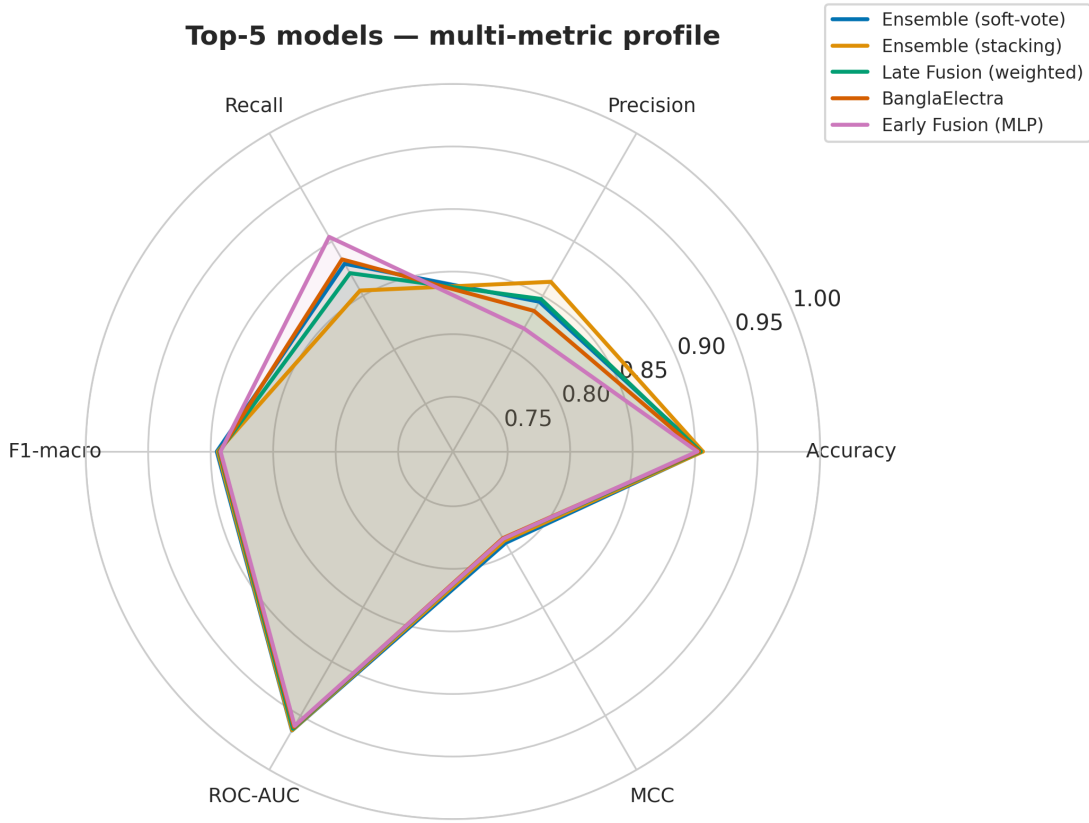


Figure 4.8: Multi-metric radar chart, top-5 models (Accuracy, Precision, Recall, F1-macro, ROC-AUC, MCC).

or two lexical triggers a difference relevant to how each architecture generalises to bullying language that lacks an isolated profane token.

4.6 Uncertainty Quantification and Calibration

4.6.1 MC-Dropout: Aleatoric vs. Epistemic Uncertainty

Twenty stochastic forward passes with dropout active at inference time (MC-Dropout [15]) give a predictive distribution per test example, decomposed into aleatoric (expected entropy irreducible, data-inherent ambiguity) and epistemic (mutual information / BALD reducible, model-inherent) components. Table 4.5 shows that misclassified examples carry more than double the predictive entropy of correctly classified ones (0.476 vs. 0.208), and both the aleatoric and epistemic components are elevated i.e., errors concentrate on genuinely ambiguous posts rather than being uniformly distributed. Figure 4.15 visualises this separation directly.

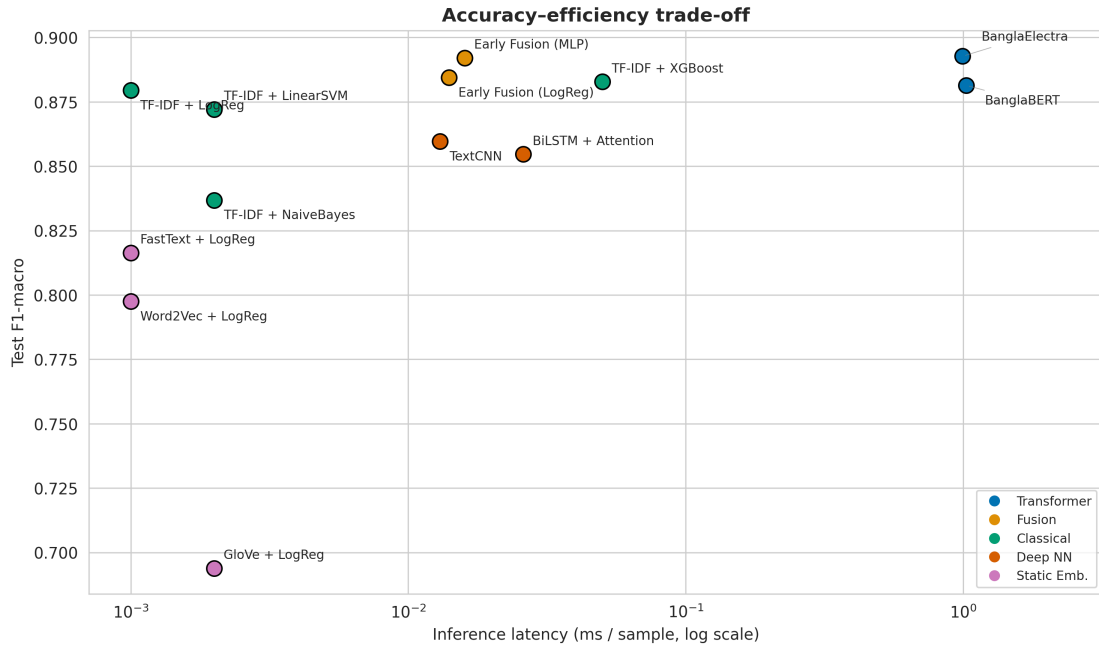


Figure 4.9: Accuracy (F1-macro) vs. per-sample inference latency, coloured by model family.

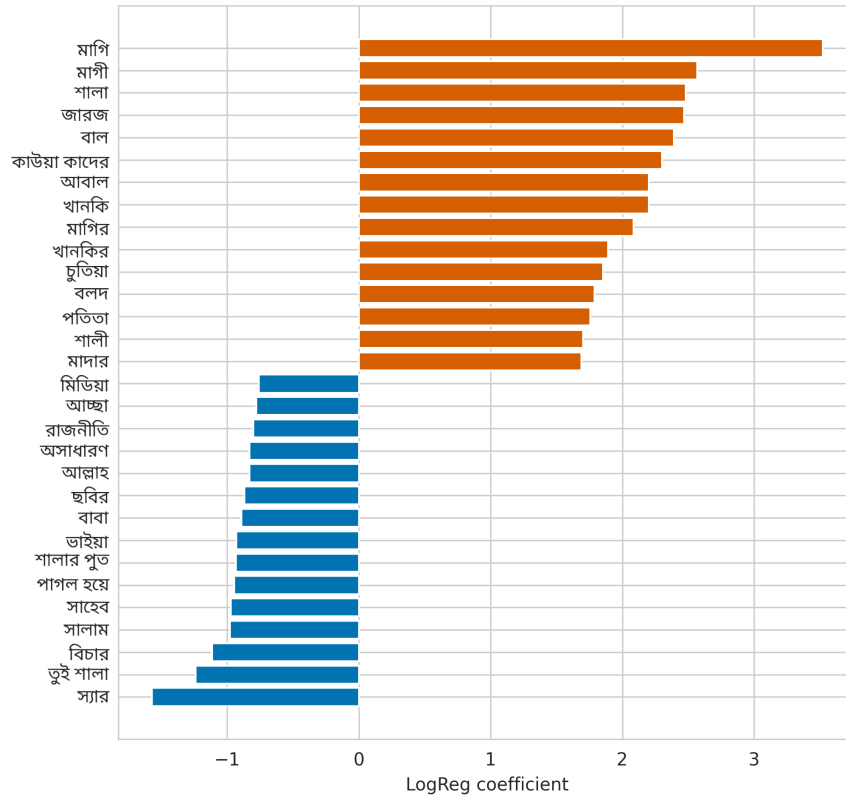


Figure 4.10: Global feature importance: top positive/negative logistic-regression coefficients over the TF-IDF space.

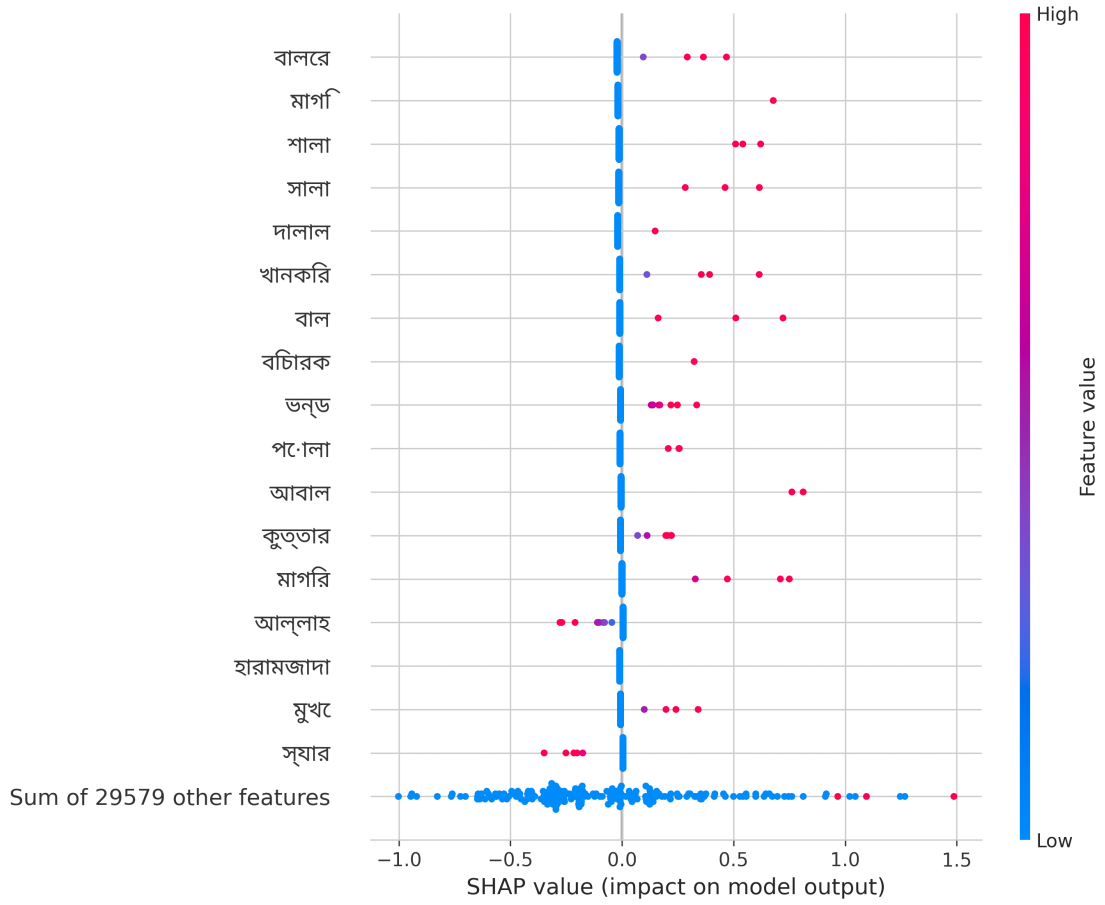


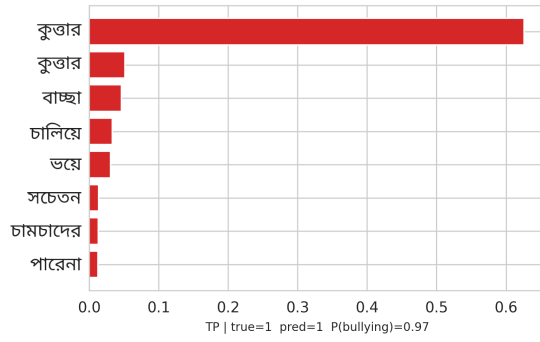
Figure 4.11: SHAP beeswarm plot for the TF-IDF logistic-regression model, top features by mean $|\text{SHAP value}|$.

4.6.2 Selective Prediction (Risk–Coverage)

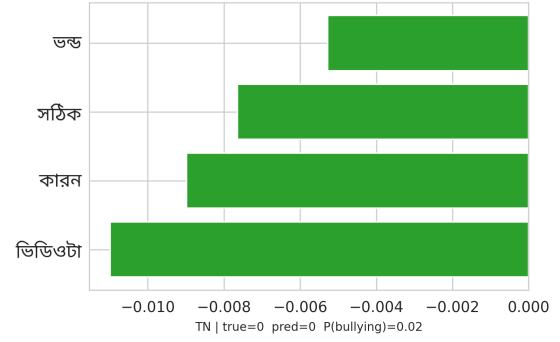
Ranking test examples by predictive uncertainty and abstaining on the least-confident fraction traces a risk–coverage curve (Figure 4.16). Three independent uncertainty signals are compared: the best transformer’s raw softmax confidence, its MC-Dropout predictive entropy, and a deep-ensemble disagreement score computed as the variance across four independently trained members (TextCNN, BiLSTM+Attention, BanglaBERT, BanglaELECTRA). All three curves show the same qualitative pattern at 80% coverage (abstaining on the most

Table 4.5: Predictive Entropy Decomposition, Correct vs. Misclassified ($n = 4,475$)

Group	n	Entropy	Aleatoric	Epistemic
Correct	4,024	0.2081	0.2037	0.0044
Misclassified	451	0.4761	0.4567	0.0194

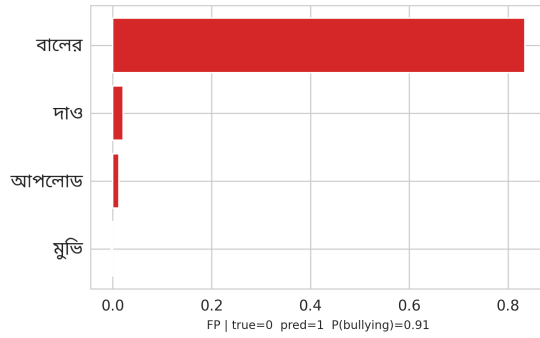


(a) True-positive prediction

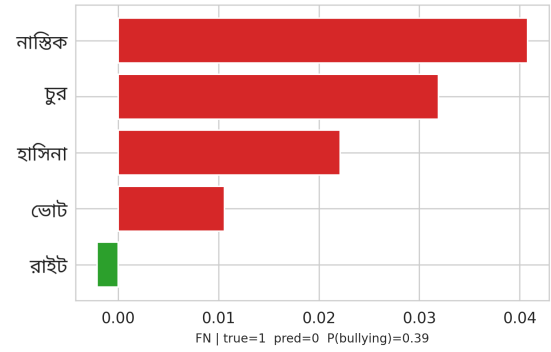


(b) True-negative prediction

Figure 4.12: LIME explanations for a true-positive (left) and true-negative (right) prediction from the best transformer.

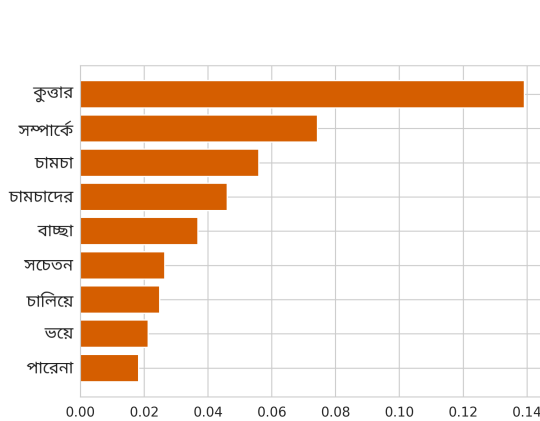


(a) False-positive prediction

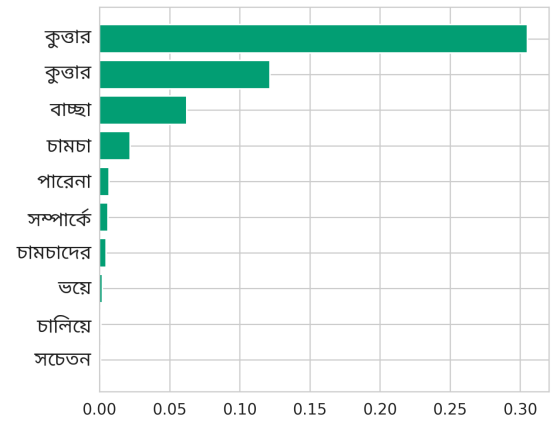


(b) False-negative prediction

Figure 4.13: LIME explanations for a false-positive (left) and false-negative (right) prediction.



(a) Transformer CLS-attention



(b) BiLSTM additive attention

Figure 4.14: Attention weights on the same test post: transformer CLS-attention (left) vs. BiLSTM additive attention (right).

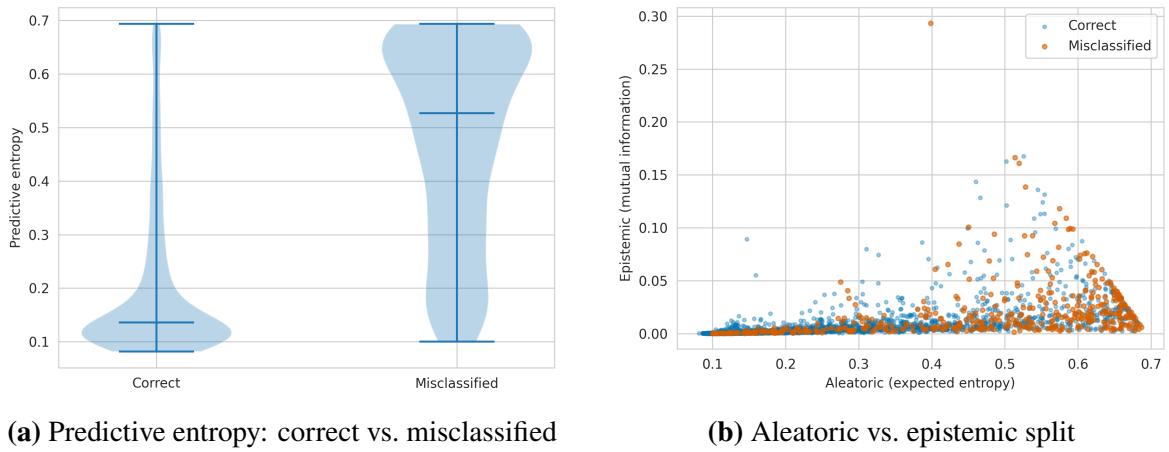


Figure 4.15: Predictive entropy for correct vs. misclassified test examples (left) and the aleatoric/epistemic split (right).

uncertain 20% of posts), test error drops substantially below the full-coverage error indicating each uncertainty estimate is well-ordered and any of the three could support a human-in-the-loop moderation queue for the least-confident cases; the area under each risk-coverage curve (AURC) gives a single-number comparison of how efficiently each signal concentrates errors into the abstained region.

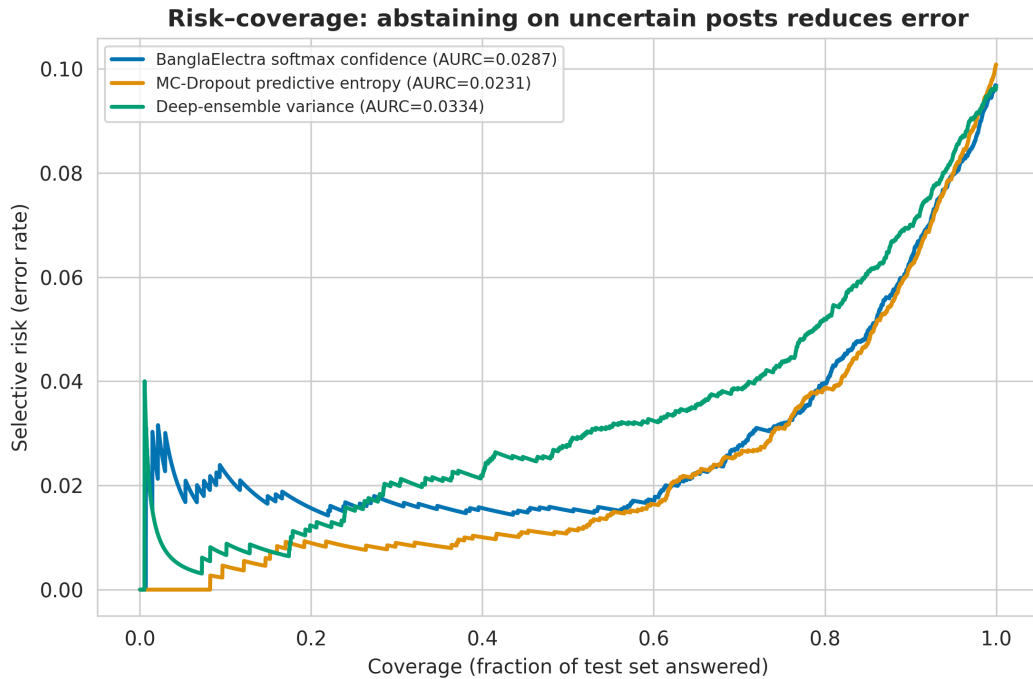


Figure 4.16: Risk-coverage curve for selective prediction on the best transformer.

4.6.3 Calibration and Temperature Scaling

Reliability diagrams before and after post-hoc temperature scaling (fitted on the validation logits, optimal $T = 1.15$) are shown in Figure 4.17, together with the calibration metrics in Table 4.6. Temperature scaling reduces ECE by 23% ($0.0324 \rightarrow 0.0248$) and MCE by 20% ($0.1724 \rightarrow 0.1380$) with a small Brier-score improvement, at zero cost to the underlying accuracy ranking (temperature scaling is a monotonic transform of the logits).

Table 4.6: Calibration Metrics Before / After Temperature Scaling

Setting	ECE	MCE	Brier	NLL
Before ($T = 1.00$)	0.0324	0.1724	0.0748	0.2563
After ($T = 1.15$)	0.0248	0.1380	0.0734	0.2505

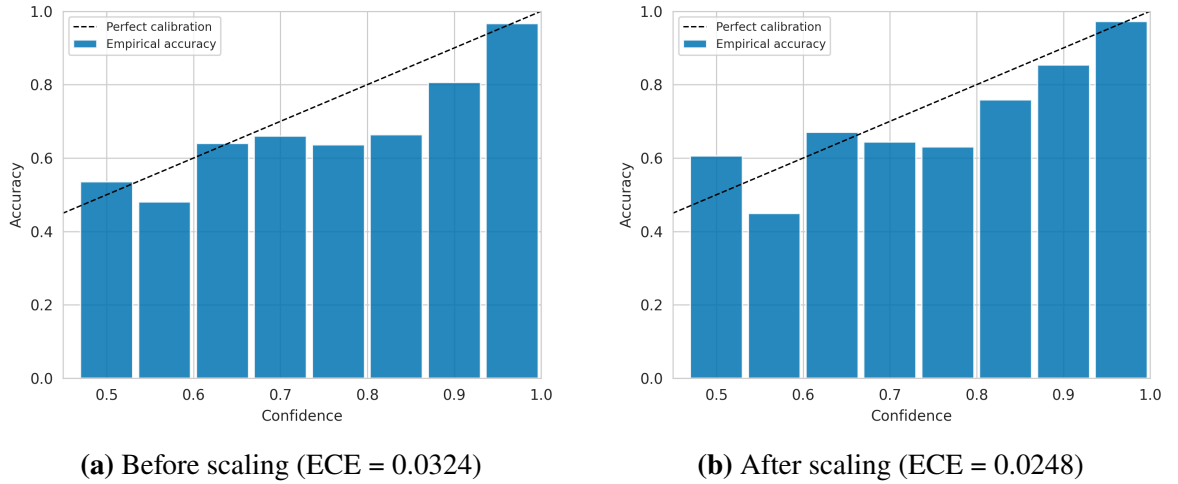


Figure 4.17: Reliability diagrams before (left, ECE = 0.0324) and after (right, ECE = 0.0248) temperature scaling.

4.7 Data Audit and Ablation Studies

4.7.1 Data Audit

Because the best models plateau around F1-macro ≈ 0.89 – 0.90 , the label parser and cleaning funnel were first audited to rule out duplicate leakage as an explanation. Table 4.7 shows label parsing produces zero NaN failures, and de-duplication removes a near-identical share of positives (0.4%) and negatives (0.6%), leaving the class ratio essentially unchanged ($33.33\% \rightarrow 33.38\%$). This confirms the performance ceiling reflects genuine task difficulty and annotation-boundary ambiguity rather than a data-pipeline defect.

Table 4.7: Data-Cleaning Funnel (Label Audit)

Stage	Rows	Pos (1)	Neg (0)	Pos share
Raw	30,000	–	–	–
Parsed labels	30,000	10,000	20,000	33.33%
After NA/empty drop	30,000	10,000	20,000	33.33%
After de-duplication	29,836	9,958	19,878	33.38%

A further 7 posts whose text normalises to an empty string under the Bengali-script cleaning filter of Section 3.3 are excluded immediately before the train/validation/test split, leaving 29,829 posts entering the stratified split reported in Table 3.3.

4.7.2 Bag-of-Words vs. TF-IDF Weighting

Holding the n-gram range fixed (word 1–2 gram char 3–5 gram), Table 4.8 isolates the effect of IDF re-weighting and L2 normalisation against raw bag-of-words counts. TF-IDF improves LogReg and LinearSVM by 1.3–1.5 F1-macro points, while count-based Naive Bayes and XGBoost are marginally better with raw counts consistent with Naive Bayes’ multinomial count assumption and tree-based models’ relative insensitivity to feature scaling.

Table 4.8: Bag-of-Words vs. TF-IDF, at Tuned Decision Threshold

Model	Rep.	F1m@0.5	F1m@tuned	Δ F1m
LogReg	TF-IDF	0.8852	0.8829	–0.0022
LogReg	BoW	0.8701	0.8696	–0.0005
LinearSVM	TF-IDF	0.8781	0.8739	–0.0042
LinearSVM	BoW	0.8503	0.8590	+0.0088
NaiveBayes	TF-IDF	0.8342	0.8409	+0.0066
NaiveBayes	BoW	0.8132	0.8448	+0.0317
XGBoost	TF-IDF	0.8793	0.8789	–0.0004
XGBoost	BoW	0.8856	0.8852	–0.0004

4.7.3 Threshold Tuning

Sweeping the decision threshold on the validation split to maximise F1-macro yields model-dependent gains: Naive Bayes benefits most (+0.019 average F1-macro across representations), linear SVM benefits modestly (+0.002), while XGBoost and logistic regression are essentially threshold-insensitive at their validation optimum (≈ -0.001). This indicates the

default 0.5 threshold is already close to optimal for the strongest models but is a cheap, retraining-free lever for weaker calibrated classifiers.

4.7.4 Auxiliary Category Feature

Concatenating a one-hot post-category vector onto the TF-IDF space changes logistic regression’s F1-macro by 0.0012 i.e., knowing the source category (sports, politics, religion, ...) does not meaningfully help a lexical classifier once the text itself is observed, suggesting the category variable’s signal is already implicit in the vocabulary distribution captured by TF-IDF.

4.7.5 Stronger Transformer Configuration

As a check on whether the transformer results reflect an under-trained configuration, both Bangla transformers were re-trained with a longer maximum sequence length ($64 \rightarrow 128$ subword tokens), more epochs ($3 \rightarrow 5$, with early stopping), and a lower learning rate ($2 \times 10^{-5} \rightarrow 1.5 \times 10^{-5}$). The re-trained models did not improve on the original configuration: BanglaBERT (v2) reached F1-macro = 0.8744 versus 0.8814 for the original three-epoch, 64-token model, and BanglaELECTRA (v2) reached 0.8764 versus 0.8927 originally both changes under one F1-macro point and, for BanglaELECTRA, slightly worse. Given the corpus’s median post length of 9 words (Table 3.1), extending the maximum sequence length to 128 tokens provides essentially no additional context for the overwhelming majority of posts, while the extra training epochs offer no compensating benefit at this corpus size; the shorter, cheaper original configuration is therefore retained as the reported transformer setting.

4.8 Generalization to a Second Corpus (BD-SHS)

4.8.1 Results

Following the replication protocol of Section 3.9, sixteen of the seventeen registered architectures were re-trained and re-evaluated on BD-SHS’s own 35,184/7,540/7,540 split. The stacked ensemble again achieves the best held-out performance (F1-macro = 0.9197, accuracy = 0.9198, MCC = 0.839, 95% CI [0.9133, 0.9260]), ahead of BanglaELECTRA (F1-macro = 0.9168) and the strongest classical baseline, TF-IDF+XGBoost (F1-macro = 0.9038). Table

4.9 reports the top and bottom models by F1-macro; the complete sixteen-model. Absolute F1-macro scores are higher on BD-SHS for many of the shared models, with the largest gains observed among the top-performing models. plausibly reflecting BD-SHS’s larger training split (35,184 vs. approximately 20,900) and more balanced label distribution, although other differences between the two corpora’s collection and annotation processes may also contribute, so absolute F1 should not be compared across corpora as if the task were identical.

Table 4.9: BD-SHS Results, Test Set $n = 7,540$ (Top/Bottom; Full Table in Repo)

Model	Family	F1-macro	ROC-AUC
Ensemble (stacking)	Ensemble	0.9197	0.9734
BanglaELECTRA	Transformer	0.9168	0.9734
TF-IDF+XGBoost	Classical	0.9038	0.9611
TextCNN	Deep NN	0.8876	0.9523
FastText+LogReg	Static Emb.	0.8289	0.9063
GloVe+LogReg	Static Emb.	0.7003	0.7709

4.8.2 Rank Stability Across Corpora

Absolute score differences aside, the more scientifically meaningful question is whether the same architectures win in the same relative order on both corpora. Ranking all sixteen shared models by F1-macro on each corpus and computing the Spearman rank correlation between the two rankings gives $\rho = 0.988$ ($p < 0.001$; Figure 4.18): the family-level ordering observed in Chapter 4 ensembles > fusion > transformers > classical TF-IDF > deep networks > static embeddings, with GloVe consistently weakest reproduces almost exactly, with only minor local swaps (BanglaELECTRA and weighted late fusion trade third/fourth place; BiLSTM and TextCNN trade eleventh/twelfth place). This directly answers the single-corpus limitation raised in Section 1.7: the comparative conclusions of this thesis are not an artifact of the particular corpus used, even though absolute performance does not transfer one-to-one between corpora of differing size and label balance.

4.8.3 Uncertainty and Calibration Also Replicate

The uncertainty and calibration findings of Section 4.6 were also checked on BD-SHS for the best transformer. Misclassified test examples again carry substantially higher predictive entropy than correctly classified ones roughly $2.9\times$ higher mirroring the correct-vs-misclassified

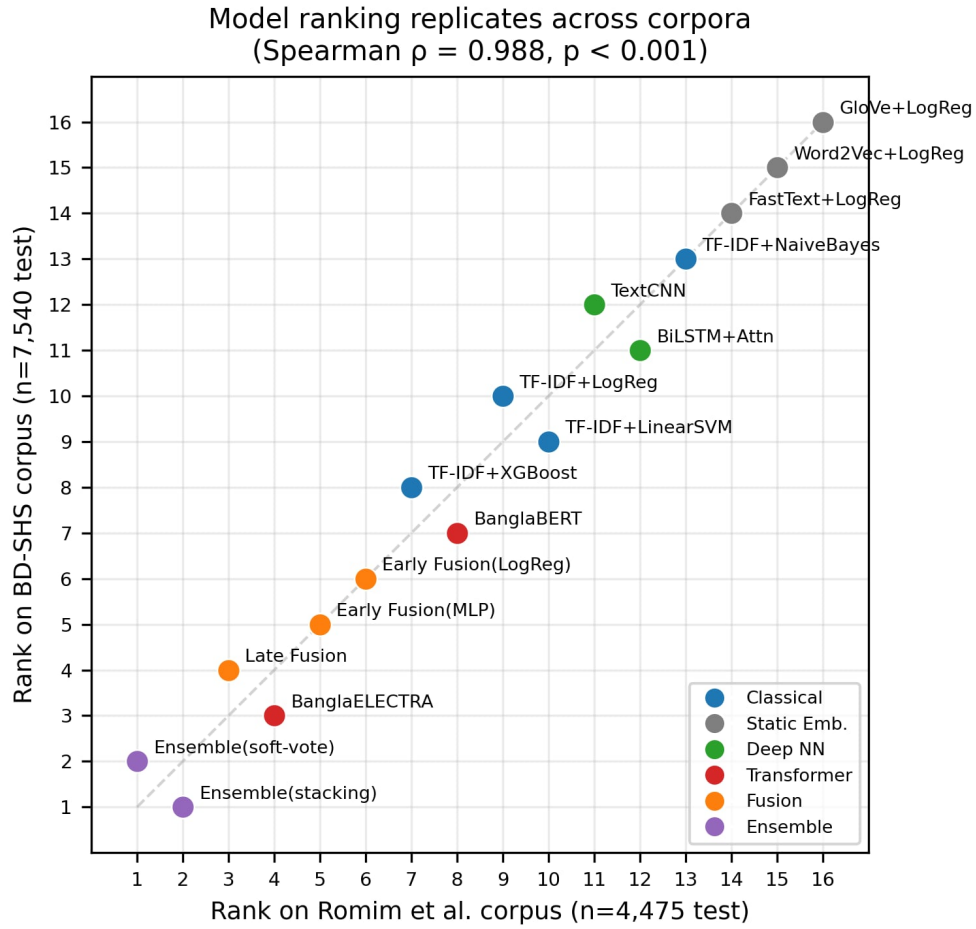


Figure 4.18: Model rank on the Romim et al. corpus vs. BD-SHS corpus; Spearman $\rho = 0.988$ ($p < 0.001$).

entropy gap reported in Table 4.4 for the primary corpus. Post-hoc temperature scaling ($T = 1.28$, fit on BD-SHS’s own validation logits) reduces Expected Calibration Error by 62% ($0.0326 \rightarrow 0.0125$), an even larger relative improvement than the 23% reduction observed on the primary corpus, consistent with both corpora’s residual errors concentrating on genuinely ambiguous posts rather than reflecting an annotation quirk specific to one dataset.

4.9 Threats to Validity

Internal validity: the reported metrics rely on a single fixed 70/15/15 stratified split; Section 4.3’s 5-fold cross-validation check on the classical models mitigates but does not eliminate the risk that this particular split is unusually favourable or unfavourable for any one

model. Construct validity: the corpus’s binary bullying/non-bullying label is a majority-vote crowd-annotation over inherently subjective judgements about what counts as “bullying”; annotation-boundary disagreement is a plausible explanation for the residual F1-macro gap discussed. External validity: because the corpus is drawn from two specific platforms (YouTube and Facebook comment sections) and seven specific topical categories, the reported results may not transfer directly to other Bengali platforms, dialectal variants, or heavily code-mixed Banglish text. Conclusion validity: the McNemar and bootstrap-CI tests in Section 4.2 address whether observed metric differences are distinguishable from sampling noise, but multiple-comparison correction across all seventeen pairwise model comparisons was not applied, so the handful of borderline significant results (p close to 0.05) should be read with appropriate caution.

4.10 Discussion

The results across Sections 4.1–4.7 indicate that:

- Fine-tuned Bengali transformers, and especially BanglaELECTRA, are strong standalone performers that already close most of the gap to the best possible ensemble.
- Fusion and ensembling deliver a real but incremental improvement, statistically distinguishable from the strongest classical baseline but not from each other.
- Explainability analysis shows the models rely substantially on explicit lexical/profanity cues, which is reassuring for interpretability but also suggests headroom for detecting more implicit or sarcastic bullying
- Predictive uncertainty is well-behaved errors concentrate on genuinely ambiguous posts, and temperature scaling cheaply improves calibration which together support a practical deployment pattern in which low-confidence predictions are routed to human moderators rather than auto-actioned.

Comparing these findings against the reproduced state-of-the-art baseline discussed in Section 2.1 and against Ashraf et al.’s cyber-troll behaviour-detection results [18] ($F1 = 87\text{--}88\%$ on a smaller, single-track English-language benchmark), the multi-track Bengali system developed in this thesis reaches a comparable or higher F1-macro (0.8951) while additionally providing calibrated uncertainty estimates and multi-view explanations that neither of those

earlier systems reported evidence that the controlled, multi-track methodology adopted here yields not just a stronger number, but a more trustworthy and auditable system overall.

These findings have important implications for the development of Bengali online-content moderation systems. The results suggest that effective moderation should not rely solely on a single high-performing model; instead, predictive performance, explainability, uncertainty, and calibration should be considered together when developing trustworthy automated systems. In practical Bengali social-media environments, such an approach could help identify potentially harmful content while allowing uncertain predictions to be reviewed more carefully by human moderators. Thus, the study connects the technical objective of improving classification performance with the broader motivation of supporting safer and more responsible Bengali online communities.

4.11 Implications for Deployment

The results in this chapter suggest a concrete, tiered deployment pattern rather than a single monolithic classifier. At the first tier, a single fine-tuned transformer (BanglaELECTRA) already delivers $F1\text{-macro} = 0.893$ at roughly 1 ms per-sample GPU inference, which is likely sufficient for most moderation workloads on its own and avoids the added engineering complexity of maintaining six separate models for an ensemble. At a second tier, where the small additional accuracy of the stacked ensemble ($F1\text{-macro} = 0.895$) is worth the added inference cost for example, in a review queue that only needs to process a smaller volume of already-flagged content the full six-member stack can be applied. Across both tiers, the calibration results of Section 4.6.3 support setting a confidence threshold (informed by the reliability diagram in Figure 4.17) below which predictions are routed to a human moderator rather than auto-actioned, and the risk–coverage analysis of Section 4.6.2 indicates that abstaining on roughly the least-confident 20% of posts would remove a disproportionate share of the model’s errors from the automated decision path. Finally, the explainability outputs of Section 4.5 particularly the LIME case studies could be surfaced directly to human moderators alongside a flagged post, giving them the specific tokens the model weighted most heavily rather than a bare probability score, which is likely to improve moderator trust and review speed relative to an unexplained black-box score.

In the absence of deployment-specific operational constraints, the proposed tiered moderation workflow should be interpreted as a conceptual framework rather than a fixed production policy. In a real deployment, decision thresholds would need to be selected according to factors such as acceptable latency, human-review capacity, and the relative costs of false-negative and false-positive decisions.

4.12 Objectives Achieved

Revisiting the five objectives stated in Section 1.3:

- The controlled multi-track benchmark was built and reported in Table 4.1, spanning all four representation families plus fusion/ensembling.
- Representation combination was investigated in Sections 4.1 and 4.2, where the stacked ensemble and weighted late fusion were both found to improve on the best individually trained model (BanglaELECTRA) by a small but statistically distinguishable margin.
- Explainability was evaluated using global (SHAP, linear coefficients), local (LIME), and mechanistic (attention-comparison) methods in Section 4.5.
- Predictive uncertainty and calibration were quantified in Section 4.6 using MC-Dropout, deep-ensemble disagreement, risk–coverage analysis, and temperature scaling.
- The dataset and key modelling choices were audited and ablated in Section 4.7, confirming that the observed performance ceiling reflects genuine task difficulty rather than a pipeline artefact.

All five stated objectives are therefore judged to have been met.

4.13 Financial Analysis and Budget

This thesis is a computational study with no physical prototype, so its direct costs are dominated by compute rather than hardware or materials. All experiments were run on the free tier of a cloud notebook platform (Kaggle, TESLA T4 GPU accelerator; Section 3.6), so the marginal compute cost of the reported results was zero at the point of use. The main resource actually consumed was wall-clock time: approximately 3 hours for the full pipeline (all five tracks, all seventeen models, all figures and tables).

4.14 Chapter Summary

This chapter reported the master comparative results across seventeen models, established the statistical significance of the fusion/ensemble improvement via bootstrap confidence intervals and McNemar tests, cross-checked the single-split results against 5-fold cross-validation, presented training curves, ROC/PR curves, a confusion matrix, a multi-metric radar, and an accuracy-latency trade-off, analysed the models' global and local explanations and their attention behaviour, quantified predictive uncertainty and calibration, audited the data pipeline and ran four ablations, and discussed the implications of these findings for deployment. The final chapter summarises these findings, states the limitations of the study, and outlines directions for future work.

Chapter V

Conclusion and Future Work

5.1 Summary of Findings

This thesis presented a controlled, multi-track comparative study of Bengali cyberbullying and hate-speech detection spanning classical lexical models, static embeddings, deep neural networks, fine-tuned transformers, and representation/decision-level fusion and ensembling, evaluated under one protocol with eleven metrics, bootstrap confidence intervals, and McNemar significance testing. A stacked ensemble achieved the best held-out performance (F1-macro = 0.895, accuracy = 0.907), narrowly but statistically distinguishably ahead of the strongest single transformer, BanglaELECTRA (F1-macro = 0.893), and clearly ahead of the strongest classical baseline, TF-IDF+XGBoost (F1-macro = 0.883). Complementary explainability and uncertainty-quantification analyses showed that (i) character n-grams and explicit lexical cues remain the dominant global signal even for transformer-era classifiers, (ii) prediction errors concentrate on genuinely high-entropy, ambiguous posts rather than being randomly distributed, and (iii) simple post-hoc temperature scaling meaningfully improves calibration at no accuracy cost. A data audit confirmed the observed performance ceiling reflects task difficulty rather than pipeline artefacts.

5.2 Limitations

- Single primary corpus, single language variety: results are anchored on one publicly available Bengali corpus [1] as the primary benchmark; although the full comparative protocol is replicated on a second, independent corpus (BD-SHS [4]; Section 3.9, Section 4.8) and the model ranking is found to reproduce almost exactly (Spearman = 0.988, $p < 0.001$), generalisation to code mixed Banglish, regional dialects, and other domains remains untested.
- Binary labelling: the corpus provides a binary bullying/non-bullying label; finer-grained categories (e.g., misogynistic, xenophobic, political) available in newer corpora [5] would allow multi-label evaluation.

- Ensemble/fusion gains are statistically modest: confidence intervals for the top four models overlap substantially, so the practical case for ensembling over a single well-tuned transformer should weigh the added inference cost and complexity against the incremental F1-macro gain (0.20.4 points).
- Explainability outputs (LIME, SHAP, attention) are illustrative case studies rather than a systematic human evaluation of explanation quality or faithfulness.

5.3 Future Work

Future research should focus on the following priorities:

- Extending the comparison to multi-label, multi-category Bengali hate-speech corpora such as HateCorpBN-XL [5] (BD-SHS [4] itself is no longer future work: Section 3.9 and Section 4.8 report a full replication of the comparative protocol on it).
- Evaluating code-mixed Banglish text explicitly.
- Incorporating parameter-efficient fine-tuning (LoRA/QLoRA) of larger multilingual LLMs, which recent work suggests is competitive for low-resource Bengali hate-speech detection.
- Running a human evaluation of the LIME/SHAP explanations for moderator trust and usability.
- Extending the behaviour-detection framing of Ashraf et al. [18] by adding multi-class behavioural labels (e.g., aggressive, frustrated, abusive) rather than a single bullying/non-bullying label, to more directly support crime-prevention-style shortlisting.

5.4 Concluding Remarks

This thesis represents a comprehensive effort to investigate Bengali cyberbullying and hate-speech detection through the comparative evaluation of multiple modelling approaches. Beyond predictive performance, the study emphasizes the importance of explainability, uncertainty analysis, and responsible evaluation when developing trustworthy language-processing systems. Through this research, we gained practical experience in data preparation, model development, experimental evaluation, and the interpretation of machine-learning results. The

project also helped us understand that developing an effective automated detection system involves not only achieving strong performance but also considering the reliability and practical limitations of model predictions. We hope that the findings of this study will contribute to future research toward safer and more responsible Bengali online environments.

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